Speaking in Code: Surveillance and Coordinated Dissent in a Language-Based Global Game with LLM Agents

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Abstract

Bad news alone does not produce collective action: citizens must also believe that others can read it as a reason to move. I study that problem in a language-based Morris–Shin regime-change game, where large language model (LLM) agents play citizens deciding whether to join an uprising. Each agent receives a private natural-language briefing generated from a noisy signal, so evidence about the state and cues about its actionability arrive as prose rather than as the canonical scalar signal. The agents reproduce the model's threshold logic, joining after weak-regime signals and staying after strong-regime ones, so participation declines with regime strength (mean $r(J, A(\theta)) = +0.80$ across 7 models). Pre-play communication preserves this threshold structure and modestly raises participation in the pooled matched-cell comparison ($\Delta = +2.10$ pp), with heterogeneous model effects. I then introduce surveillance only where peer messages are written, leaving the receiver's decision prompt unchanged. Point estimates are negative in 5 of 5 models, with 4 individually significant (equal-weighted mean $\Delta = -11.2$ pp), and a nested 500-cell replication on a second, publicly accessible model confirms the effect ($\Delta = -8.0$ pp). A codedness control that elicits even more heavily coded messages without any monitoring framing leaves participation nearly unchanged ($\Delta = -0.7$ pp), so the effect is not generic stylistic vagueness. In elicitations that do not re-show peer messages, stated expectations about others' participation remain stable while actions fall, and surveilled messages shift from direct political terms toward coded phrases. Surveillance need not hide bad news or enter the receiver's choice directly: it suppresses action through what guarded messages stop conveying about others' willingness to act.

Short abstract (100 words). Collective action requires more than bad news: citizens must know that others can read it as a reason to move. I study this in a language-based Morris–Shin regime-change game with LLM citizens receiving noisy natural-language briefings and deciding whether to join an uprising. Seven models reproduce threshold behavior. Under sender-side surveillance, joining falls in all five prompt-isolation models (four significant) and in a nested 500-cell replication, while stated expectations stay flat in restricted elicitations. A codedness control—more heavily coded messages without monitoring framing—leaves joining nearly unchanged: surveillance suppresses action through what guarded messages convey, not coded form alone.

JEL: C72, C92, D82, D83, P16
Keywords: global games, regime change, surveillance, communication, LLM agents, self-censorship

1 Introduction

In a coordination crisis, information matters only insofar as it is *usable*. A citizen deciding whether to join an uprising needs a view of the regime's strength, and also of whether other citizens are prepared to act on what they know. In the canonical global game of regime change (Morris and Shin, 2003), private signals alone resolve equilibrium multiplicity (Carlsson and van Damme, 1993; Frankel et al., 2003). Real coordination crises, from bank runs (Diamond and Dybvig, 1983) to currency attacks (Morris and Shin, 1998; Obstfeld, 1996) to political upheaval (Angeletos et al., 2007), add a social layer: people talk before they move. A message in that layer has two roles. It reports a state of the world, and it reveals something about the speaker's willingness to make that report openly. That dual role is what makes communication valuable, and also what makes it vulnerable to surveillance.

The logic of the global game is simple. A regime has some underlying strength, but no citizen knows it exactly. Each citizen receives a private, noisy signal—some see evidence of weakness, others see evidence of stability—and must decide whether to join an uprising or stay home. Joining pays only if enough others join too. Staying home is safe, but forgoes the chance of change. The strategic tension is that each citizen's best action depends on what they believe *others* will do, which in turn depends on what others believe, and so on. Carlsson and van Damme (1993) and Morris and Shin (2003) showed that when citizens receive sufficiently precise private signals, this regress collapses to a unique threshold equilibrium: above the threshold citizens stay; below it, they join. The theory is attractive because it turns coordination failure into a setting with sharp comparative statics.

Testing these predictions empirically is difficult. Laboratory experiments confirm threshold behavior with stylized numeric signals (Heinemann et al., 2004, 2009; Shurchkov, 2013; Szkup and Trevino, 2020; Helland et al., 2021), but they are less suited to manipulating the rich, multi-source information that characterizes real crises. Field evidence from actual coordination failures confounds strategic behavior with institutional differences, repression risk, and unobserved information structures. The same problem is especially acute for surveillance. When dissent falls after monitoring expands, one cannot tell whether citizens are personally deterred from acting or whether peers have begun to speak in ways that no longer help others coordinate.

In this paper, I ask whether the second channel can operate on its own. The mechanism I test works before the moment of action, by distorting the language environment itself: does monitoring change what citizens say, and are those changed messages enough to move receivers' actions? Within the experiment, the answer to both questions is yes, and the movement is not reproduced by coded style alone. The receiver never sees a surveillance warning in the main treatment; only the sender's message-generation environment changes. I do not claim that surveillance leaves all beliefs unchanged; the argument is narrower. Authoritarian control can operate on what might be called the conversion margin, where bad news remains present in the language environment but becomes less public and less usable as a shared reason to act.

The empirical strategy treats LLM agents as a measurement instrument for a language information structure, not as a direct substitute for human subjects. I embed 7 models spanning 6 architecture families in the Morris and Shin (2003) game. Each agent receives a private signal $x_i = \theta + \varepsilon_i$ rendered as a natural-language intelligence briefing whose content shifts monotonically with signal strength across eight evidence domains. No explicit payoff table is provided; the stakes are embedded in the narrative, so agents must form action-relevant beliefs from language. The key design move is then to hold the latent state, the private briefings, the receiver's decision prompt, and the decision rule fixed while changing only the incentives under which peer messages are produced. I establish the behavioral benchmark first and introduce the surveillance treatment only after threshold-like signal use has been verified.

The paper makes three contributions. First, I give the canonical global game a richer language implementation and verify that the benchmark comparative static survives the move from numeric signals to prose: across 1,800 period-level observations and 7 models, participation is strongly decreasing in regime strength and aligned with the benchmark attack mass (mean $r = +0.80$, with $p < 0.001$ for every model). Second, on that benchmark I isolate a communication channel that field data cannot easily separate: pre-play messages carry regime-strength information and affect decisions, and in the pooled matched-cell comparison communication raises participation by +2.10 pp, though model-specific participation effects are heterogeneous. Third, and centrally, I find that consequential monitoring depresses participation even when it is introduced only at the message-generation stage. In the primary model, and in a fully nested 500-cell replication on a second model family, changing only the incentives under which peers write messages reduces joining, with the receiver's decision prompt held bit-identical. The channel is surveillance-specific: a codedness control that elicits even more heavily coded messages without monitoring framing leaves joining nearly unchanged, so the effect runs through what surveilled messages convey rather than through indirect form as such. When peer messages are excluded from the belief prompt, reported expectations about how many other citizens will join remain stable while actions fall, both post-decision in the primary model and pre-decision in a nested replication. Under a messages-included elicitation, beliefs do move, so the wedge is a fact about the restricted information set rather than a mediation result.

The mechanism evidence narrows what changed. Message controls show that decision-stage agents use peer messages as part of their information environment. The text itself reveals that surveilled messages do not simply lose bad news: in the primary model, explicit weakness references rise, while direct references to the regime and to action fall and metaphorical, deniable phrasing rises. A discriminant test in Llama 3.3 70B swaps the coordination decision for a private bet on regime fall, and surveillance still moves that task, showing a downstream judgment shift outside the main JOIN/STAY action prompt. Because a bet on regime fall can still depend on expectations about others' participation, I do not treat the gap between tasks as a structural coordination-specific decomposition. Taken together, the evidence suggests that surveillance can leave information flowing while degrading the form in which that information becomes public coordination value.

The remainder of the paper proceeds as follows. Section 2 situates the experiment in the related literature. Section 3 presents the theoretical framework, and Section 4 describes the experimental design. Section 5 establishes a threshold-like behavioral benchmark and shows that pre-play communication creates an informative channel. Section 6 tests whether an authoritarian regime can exploit that channel through surveillance, and Section 7 concludes.

2 Related Literature

A large theoretical literature studies regime change as a coordination problem. The global game framework, from Carlsson and van Damme (1993) through Morris and Shin (2003) and the dynamic extensions of Angeletos et al. (2007) and Chassang (2010), provides the theoretical foundation for this paper, and its threshold predictions are confirmed by the laboratory experiments cited above. Within that framework, Morris and Shin (2002) show that the value of information depends on its publicness, and Cornand and Heinemann (2014) measure experimentally how subjects weight private against public information under strategic complementarities. The communication layer studied here is closest to Avoyan (2024), who studies pre-play communication in global games theoretically and experimentally; classic treatments of cheap talk and pre-play communication include Crawford and Sobel (1982), Farrell and Rabin (1996), Blume and Ortmann (2007), and Ellingsen and Östling (2010). On the manipulation side, Edmond (2013) characterizes a regime that distorts the signals citizens receive in a regime-change global game, and information-design treatments of coordination include Goldstein and Huang (2016), Inostroza and Pavan (2025), and Kolotilin et al. (2022) on censorship as optimal persuasion. The design here differs from that work in who is being manipulated. The regime does not choose the receiver's signal structure; surveillance changes the incentives under which *peers* write messages, and the question is what the resulting message distribution does to the receiver's coordination problem.

The political economy of authoritarian control provides the substantive motivation. Kuran (1991) models preference falsification under social pressure, Shadmehr and Bernhardt (2011) characterize the collective action problem under repression, Guriev and Treisman (2019) describe informational autocrats who rule by manipulating information rather than by mass violence, and Little (2016) models how communication technology enters protest. Empirically, King et al. (2013) document that Chinese censorship targets posts about collective action specifically, suggesting that authorities perceive the coordination channel as the binding threat; Chen and Yang (2019) show that low demand for uncensored information and limited social transmission can make censorship robust; Roberts (2018) documents friction-based censorship; and Xu (2021) studies digital surveillance as an instrument of preemptive control. Penney (2016) and Stoycheff (2016) show that surveillance announcements suppress online expression, while Enikolopov et al. (2020) show that social media facilitates protest. The paper's central object, whether others' willingness to act remains legible, is the second-order-belief margin that Bursztyn et al. (2020b) and Bursztyn et al. (2020a) measure in human populations, and that Cantoni et al. (2019) study for protest participation under strategic complementarity. These literatures establish a robust empirical fact: surveillance and censorship reduce dissident behavior. Field data, however, rarely separate the two channels through which they could operate: *direct deterrence*, in which citizens fear punishment for participation, and *communication poisoning*, in which peers self-censor such that downstream agents lose coordination cues. Rather than re-establish that surveillance suppresses dissent, I isolate the second channel with a design that holds individual deterrence fixed and varies only the messaging environment.[1]

The framing is also adjacent to information-design work that separates belief movement from the way beliefs map into action. Bayesian persuasion treats information as a selected distribution of posterior beliefs (Kamenica and Gentzkow, 2011; Bergemann and Morris, 2019; Mathevet et al., 2020), and Gans (2026) emphasizes that actions can also be steered without changing beliefs, by changing the procedure that translates a belief into an action. The channel here is different: the receiver's decision rule is held fixed, but surveillance changes whether information becomes public and action-guiding in a coordination game.

Finally, on the methodological side, Horton (2023) proposed LLMs as "homo silicus"; Aher et al. (2023) and Argyle et al. (2023) use language models to replicate human-subject results and simulate survey samples, and Mei et al. (2024) find chatbot behavior within the human distribution in canonical games. Subsequent work found that LLMs struggle in coordination games (Akata et al., 2025; Sun et al., 2025), with model scale not predicting strategic performance (Jia et al., 2025). Ouyang et al. (2024) documents that alignment shifts risk preferences, while Cheung et al. (2025) documents amplified moral-decision biases. Those biases are exactly why the language environment deserves direct study: what matters is how agents map politically loaded prose into action, not whether they optimize from an explicit payoff table. I therefore use narrative stakes as the implemented decision environment and audit that choice with payoff-grid checks, narrative-stakes variants, and text-only classifier diagnostics. Critical reviews by Larooij and Törnberg (2025, 2026) identify validation challenges that I respond to through cross-generator robustness (Appendix E.8), scramble/flip falsification, held-out classifier diagnostics, and cross-model prompt-isolation replication.

3 The Global Game of Regime Change

A continuum of citizens indexed by $i \in [0, 1]$ simultaneously choose whether to join an uprising ($a_i = 1$) or stay home ($a_i = 0$). The regime has strength $\theta \in \mathbb{R}$, drawn from a diffuse (improper uniform) prior. States $\theta < 0$ represent regimes so weak they fall without opposition under the strict rule below; states $\theta \geq 1$ represent regimes that survive even unanimous attack. The regime falls if the mass of citizens who join exceeds θ (Eq. 1):

$$\text{Regime falls} \iff A \equiv \int_0^1 a_i \, di > \theta. \tag{1}$$

[1] Concurrent work by Ruaño and Rajan (2026) places LLM depositors in a bank-run coordination game with network contagion, finding sharp phase transitions consistent with Diamond–Dybvig.

Payoffs depend on the citizen's action and the outcome (Eq. 2):

$$u_i(a_i, A, \theta) = \begin{cases} B & \text{if } a_i = 1 \text{ and } A > \theta \\ -C & \text{if } a_i = 1 \text{ and } A \leq \theta \\ 0 & \text{if } a_i = 0 \end{cases} \tag{2}$$

where $B > 0$ is the payoff to joining a successful uprising and $C > 0$ is the cost of joining a failed attempt. Non-participants receive zero regardless of the outcome.

Each citizen observes a private signal $x_i = \theta + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ independently across citizens.

Proposition 1 (Morris and Shin, 2003)**.** *In the limit of diffuse priors, there exists a unique Bayesian Nash equilibrium (BNE) in threshold strategies. An agent joins if and only if $x_i < x^*$ (Eq. 3), where*

$$x^* = \theta^* + \sigma\Phi^{-1}(\theta^*) \tag{3}$$

$\theta^* = B/(B + C)$, and Φ denotes the standard normal CDF.

The *attack mass* $A(\theta)$, the theoretical fraction of the population that joins at regime strength θ, is given by:

$$A(\theta) = \Phi\left(\frac{x^* - \theta}{\sigma}\right). \tag{4}$$

This is a decreasing function of θ: weaker regimes face larger uprisings.

I define $J(\theta)$ as the *empirical join fraction*: the finite-sample, observed counterpart to the theoretical attack mass $A(\theta)$. I use $n = 25$ agents and a researcher-side state-generating distribution for θ; agents are not shown the distribution or (B, C, σ). I use the continuum $A(\theta)$ as an external behavioral benchmark throughout, not as the exact equilibrium of the implemented finite-n game. Throughout the paper, $r(J, A(\theta))$ denotes the Pearson correlation between the empirical join fraction J and the benchmark attack mass $A(\theta)$ across period-level observations.

The textbook model treats the information structure as exogenous: signals arrive, agents update, and actions follow. The experiment adds a pre-play language technology. Agents first map private briefings into messages, then other agents map their own briefing plus received messages into actions. Let $\mathcal{B}_i$ denote the sender's state-dependent private briefing object: the prose briefing generated from the normalized signal and the briefing generator's language context. The full LLM context also includes game instructions; the formal object $\mathcal{B}_i$ isolates the part held fixed across baseline and surveillance. Let $\mu_0(m \mid \mathcal{B}_i)$ denote the baseline distribution of messages m generated from that briefing and $\mu_S(m \mid \mathcal{B}_i)$ the distribution generated under monitoring. The surveillance treatment changes $\mu(\cdot \mid \mathcal{B}_i)$, not θ, not the private briefing, and not the receiver's decision prompt. The question is therefore whether replacing μ_0 with μ_S changes $J(\theta)$.

This replacement can matter even if messages still contain first-order information about θ. In a global game, a message carries two kinds of content: evidence about the fundamental and evidence about the publicness of that evidence. Direct terms, named targets, and open statements of weakness make a private signal more mutually legible; they help a receiver infer not only that the regime is weak, but that others are willing to treat the same fact as a reason to act. Monitoring can therefore preserve bad news while degrading this higher-order content. Coded language is the visible signature of that degradation rather than its sole cause—the codedness control and the decoding test in Section 6 measure how much of the effect travels with the form itself.

The briefing generator, the system that converts z-scores into natural-language intelligence briefings, uses three latent sliders (direction, clarity, and coordination) to select phrases across eight evidence domains. This is less an isomorphic prose transcription of the scalar Morris–Shin private signal than a controlled language environment inspired by the canonical model, with explicit state, clarity, and actionability content. Full generator details are in Appendix F.1.

4 Experimental Design

The experiment has two phases. In the first, I establish that LLM agents display threshold-like policies in the global game when private signals are conveyed in natural language, using a pure treatment (private signals only), a communication treatment (pre-play messaging), and falsification tests. In the second, I test whether an authoritarian regime can exploit the communication channel through surveillance. All LLM interactions use the same prompt structure across models.

In the theory (Section 3), θ is an unbounded fundamental. In the benchmark experiments, θ is drawn from a normal distribution and agents are not shown payoff parameters (B, C). I compute the benchmark $A(\theta)$ under the canonical normalization $B = C = 1$ ($\theta^* = 0.50$, $\sigma = 0.3$).

Throughout, $\theta^* = B/(B + C)$ is the theoretical fundamental cutoff and $x^* = \theta^* + \sigma\Phi^{-1}(\theta^*)$ is the signal cutoff. The empirical cutoff $\hat{\theta}^*$ is the 50% join point of the fitted state curve $P(\text{join} \mid \theta) = 1/(1 + e^{b_0 + b_1\theta})$, where $b_1 > 0$ corresponds to a join curve that is *decreasing* in θ (the expected direction), so $\hat{\theta}^* = -b_0/b_1$. In the canonical $B = C = 1$ benchmark, this empirical midpoint is directly comparable to the theoretical 50% point of $A(\theta)$; in payoff and signal-grid variants I report it as a fitted curve summary, not as the continuum equilibrium cutoff itself. I write $\kappa \equiv b_1$ for the logistic steepness parameter in tables, reserving β for regression coefficients. Note the sign conventions: positive κ corresponds to a join curve *decreasing* in θ, whereas in the agent-level logit regressions (Table A12) the same comparative static appears as a negative coefficient on θ.

For each country–period, a base country mean and period perturbation define z_{ct}, and nature draws $\theta \sim \mathcal{N}(z_{ct}, 1)$. Each agent i receives a private signal $x_i = \theta + \varepsilon_i$. The platform normalizes that signal to $z_i = (x_i - z_{ct})/\sigma$ and translates z_i into a multi-paragraph intelligence briefing via a deterministic generator that maps signal strength to narrative content about regime stability, economic conditions, public sentiment, and coordination prospects. This normalization is in units of private-signal noise around the country–period mean; it is not standardized by the prior-predictive signal standard deviation $\sqrt{1+\sigma^2}$. Across state and noise draws, $(x_i - z_{ct})/\sigma$ has variance $(1+\sigma^2)/\sigma^2$, so the briefing's neutral point is $x_i = z_{ct}$ rather than an absolute value of θ. The benchmark $A(\theta)$ is still evaluated at each realized period-level θ; the comparison is external in the sense that it does not use the z_{ct}-centered briefing normalization or the realized private signal draws. Agents observe the briefing, not the normalization inputs (z_{ct}, σ) themselves. Figure 1 summarizes the signal-to-text-to-decision pipeline.

Agents observe only their private briefing and never the payoff parameters (B, C), the noise level σ, or the prior distribution. The diffuse-prior equilibrium (Proposition 1) serves as an external benchmark for behavioral shape and comparative statics, not as a claim about the exact equilibrium of the implemented game.

The behavioral benchmark has four core treatments: a *pure global game* (independent decisions), *communication* (pre-play messaging on a Watts–Strogatz network, $k = 4$, $p = 0.3$), *scramble* (cross-period briefing permutation breaking the link between θ and the briefing received in that period), and *flip* (negated z-scores). A *degraded-messages* control replaces LLM messages with generic text, and a no-message control removes peer messages entirely; together they benchmark how much the decision stage depends on message content. Additional robustness tests are in Appendix E.

I test 7 models spanning 6 families (Table 1). The main benchmark and treatment runs use $n = 25$ agents per period-level task row and $\sigma = 0.3$; Appendix E varies the agent count. Benchmark statistics are reported against the canonical $B = C = 1$ attack-mass curve ($\theta^* = 0.50$); agents are not shown the payoff parameters. Main runs use temperature = 0.7 with a single sample per decision, routed through OpenRouter and cached by request hash (Appendix F); Appendix E.3 varies temperature directly. Within each matched comparison, the signal-to-briefing map is held fixed and the treatment changes only the communication environment. I report the five hypotheses examined in this paper as a single hypothesis family; the four directional results—threshold alignment, signal-direction flip, communication informativeness, and surveillance—all survive Bonferroni correction over the family at $\alpha = 0.01$, while the fifth, the scramble falsification, is a specification check rather than a directional hypothesis (Appendix Table A1). Throughout, n denotes agents per period-level task row and N denotes the number of such rows.

The unit of comparison is the matched country–period cell (θ draw plus agent-level decoding stochasticity). For agent-level regressions, standard errors are clustered at the model–country–period level. For period-level correlations, I report Fisher-transformed z confidence intervals. Because periods nest within 10 countries with persistent country means, period-level cells within a country are correlated; for the pooled benchmark OLS slope, clustering at the country level raises the standard error from 0.0100 (HC1) to 0.0375, leaving the benchmark relationship highly significant. Appendix E.4 reports the corresponding country-clustered check for the primary surveillance contrast.

The clean surveillance treatment appends a monitoring warning to the baseline communication prompt; the decision prompt is unchanged (full prompts in Appendix B). Thus the treatment is not a direct receiver warning but a manipulation of the message-generation environment whose downstream effect is measured on otherwise identical decision calls.

Two scope caveats apply to everything that follows. The first concerns training-data familiarity: the Morris–Shin game and human accounts of coded speech under surveillance (samizdat, Aesopian language) are in every model's training corpus, so both threshold behavior and the direct-to-coded message shift may partly reflect reproduction of learned patterns rather than independently derived strategy. A direct audit (Appendix D) confirms that models can recognize the setting and articulate the experimenter's hypothesis when asked. For the measurement claim this is not disqualifying, since the agents need only map language to action coherently, but it cautions against reading the behavior as agent intent, and I flag it wherever interpretation leans that way. Second, the results characterize the specific hosted model snapshots and decoding configuration listed in Table 1, not LLMs as a stable population; temperature robustness (Appendix E.3) covers the benchmark phase only, so the surveillance contrast carries no decoding-robustness check.

5 Behavioral Benchmark

Result 1 (Monotone Threshold-Like Signal Use)**.** *Across 7 models and 1,800 period-level observations in the pure global game treatment, the Pearson correlation between the empirical join fraction and the theoretical attack mass $A(\theta)$ (Eq. 4) averages $r = +0.80$ ($p < 0.001$ for every model). Because $A(\theta)$ is monotone-decreasing in θ, this correlation establishes monotone signal use with threshold-like shape, not equilibrium recovery; the empirical sigmoid is shifted leftward of the diffuse-prior cutoff (Figure 2).*[2]

[2]The primary model (Mistral) was run in multiple batches with `--append`, producing 1,000 rows from 680 unique (country, period, θ) configurations. Averaging within each configuration (deduplication) changes r from 0.753 to 0.739; all qualitative results are unaffected.

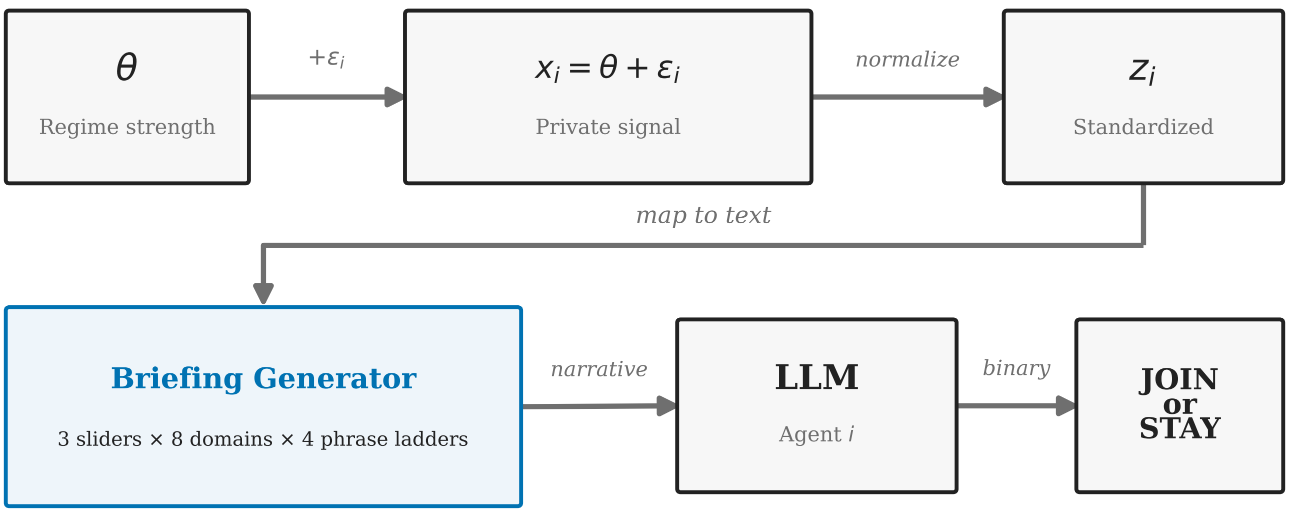

Figure 1: Signal-to-text-to-decision pipeline. Regime strength θ generates private signals x_i, converted to z-scores and rendered into natural-language briefings via eight evidence domains and three latent sliders. Conditional on the signal draw, the briefing layer is deterministic; remaining stochasticity enters through task draws and LLM decoding.

Table 1: Model summary. Columns report period-level task rows in the pure, communication, and falsification (scramble+flip) benchmark suites. These core suites use $n = 25$ agents per row and $\sigma = 0.3$; Appendix E varies agent count.

Model	Identifier (OpenRouter)	Arch.	Pure	Comm	Falsif.
Mistral Small Creative	`mistralai/mistral-small-creative`	Mistral	1000	1000	300
Llama 3.3 70B	`meta-llama/llama-3.3-70b-instruct`	Llama	100	100	200
Qwen3 30B	`qwen/qwen3-30b-a3b-instruct-2507`	Qwen (MoE)	100	100	200
GPT-OSS 120B	`openai/gpt-oss-120b`	GPT	200	200	1000
Qwen3 235B	`qwen/qwen3-235b-a22b-2507`	Qwen (MoE)	200	1000	200
Trinity Large	`arcee-ai/trinity-large-preview_free`	Arcee	100	100	200
MiniMax M2-Her	`minimax/minimax-m2-her`	MiniMax	100	100	200
Total			**1800**	**2600**	**2300**

Notes: Seven models spanning six architecture families; the identifier column gives each model's OpenRouter slug. Subsequent tables abbreviate Mistral Small Creative as Mistral, Trinity Large as Trinity, and MiniMax M2-Her as MiniMax M2. Counts are period-level task rows with 25 agents each in the core benchmark suites. Mistral includes appended batches; duplicate-cell accounting is reported in the main text and Table A7. The communication total of 2,600 counts rows as collected; 1 Trinity Large row (all 25 agent calls failing at the provider) yields no valid join fraction, so Table A7 counts 2,599 valid communication rows. Models were accessed as hosted snapshots via the OpenRouter API; archived run manifests record a February 2026 collection window, and raw request–response pairs are cached with the run outputs. Core runs use $\sigma = 0.3$ and temperature = 0.7 unless otherwise stated.

Figure 2 shows the empirical sigmoid for the primary model. Correlations span $r \in [+0.75, +0.84]$ across architectures (Table 2); the pooled OLS is $J = -0.01 + 0.64\,A(\theta)$ ($R^2 = 0.57$). Model-specific action biases shift the intercept but not the correlation (Figure A7). The shifted-grid construction in the payoff sweep does not separate BNE from level-k behavior (Szkup and Trevino, 2020); I claim threshold-like signal use, not equilibrium recovery. Per-model logistic fit parameters are reported in Appendix Table A13.

Result 2 (Signal Dependence)**.** *Cross-period scrambling of briefings reduces the mean correlation from $r = +0.80$ (raw, per-model) to $r = +0.03$ (within-country, demeaned to remove the between-country component that the within-country permutation leaves intact) across 7 models. The pooled correlation drops from $r = +0.76$ to $r = +0.01$ (Fisher $z = 26.17$, $p < 0.001$).*

Result 3 (Signal Direction)**.** *Inverting the signal direction flips the mean correlation from $r = +0.80$ to $r = -0.80$ across 7 models. The pooled correlation moves from $r = +0.76$ to $r = -0.79$ (Fisher $z = 53.85$, $p < 0.001$).*

The pure → scramble → flip pattern replicates across all 7 models (Figure 3; Table 2).

Table 2: Threshold-policy alignment by model and treatment. Cells report Pearson r between the empirical join fraction and the benchmark attack mass $A(\theta)$ under $B = C = 1$ (so $\theta^* = 0.50$); 95% Fisher-z confidence intervals in brackets.

	Main treatments		Falsification			
Model	Pure	Comm	Scramble	Flip	N (P/C/S/F)	Mean join (%)
Mistral Small Creative	+0.75 [0.72, 0.78]	+0.74 [0.71, 0.76]	−0.03 [−0.17, 0.11]	−0.81 [−0.87, −0.73]	1000/1000/200/100	38.7%
Llama 3.3 70B	+0.82 [0.74, 0.87]	+0.78 [0.69, 0.85]	+0.02 [−0.18, 0.21]	−0.84 [−0.89, −0.77]	100/100/100/100	43.9%
Qwen3 30B	+0.76 [0.66, 0.83]	+0.77 [0.68, 0.84]	+0.13 [−0.06, 0.32]	−0.87 [−0.91, −0.81]	100/100/100/100	49.9%
GPT-OSS 120B	+0.79 [0.74, 0.84]	+0.78 [0.72, 0.83]	−0.01 [−0.09, 0.08]	−0.80 [−0.83, −0.77]	200/200/500/500	40.7%
Qwen3 235B	+0.80 [0.74, 0.84]	+0.77 [0.75, 0.80]	+0.06 [−0.14, 0.25]	−0.85 [−0.90, −0.78]	200/1000/100/100	41.6%
Trinity Large	+0.84 [0.77, 0.89]	+0.83 [0.75, 0.88]	+0.05 [−0.15, 0.24]	−0.80 [−0.86, −0.71]	100/100/100/100	46.4%
MiniMax M2-Her	+0.82 [0.74, 0.87]	+0.82 [0.74, 0.87]	+0.01 [−0.19, 0.20]	−0.62 [−0.73, −0.49]	100/100/100/100	43.6%
Pooled	+0.76 [0.74, 0.78]	+0.76 [0.74, 0.77]	+0.01 [−0.04, 0.07]	−0.79 [−0.81, −0.77]	1800/2600/1200/1100	40.9%
Mean across models	+0.80	+0.78	+0.03	−0.80	—	—

Notes: $r = r(J, A(\theta))$: Pearson correlation between empirical join fraction and theoretical attack mass under $B = C = 1$, with 95% Fisher-z confidence intervals in brackets for every treatment. Join fraction uses valid decisions only (excluding parse errors); mean join is the pure-treatment average in percent. N counts period-level rows (not agents) in the pure (P), communication (C), scramble (S), and flip (F) arms. Pooled: all period-level rows concatenated; Mean: equal-weighted average across models. Duplicate-cell accounting is reported in the main-text footnote and Table A7.

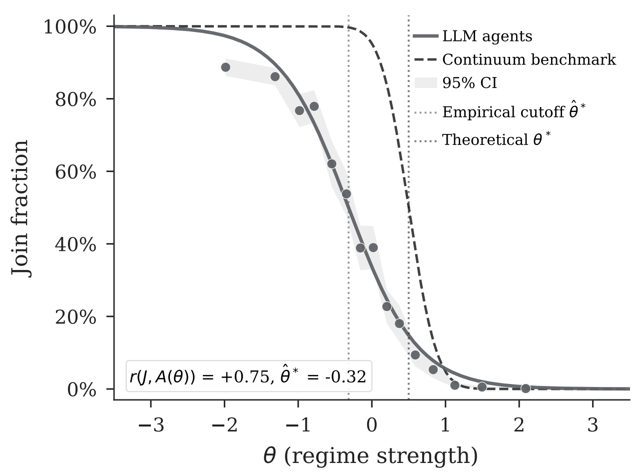

Figure 2: Empirical join fraction vs. θ (Mistral, 1,000 period-level rows, pure treatment). Dots are binned period-level join fractions; the shaded band is a 95% confidence interval around the empirical curve; dotted vertical lines mark the fitted cutoff $\hat{\theta}^* = -0.32$ and the theoretical $\theta^* = 0.50$. The empirical sigmoid is shifted leftward relative to theory. Cross-model results in Table 2 (mean $r = +0.80$, all $p < 0.001$).

5.1 Communication

Result 4 (Communication Creates an Informative Channel)**.** *Pre-play communication preserves the threshold structure and transmits regime-strength information through messages. A matched-cell comparison across* $N = 1479$ *unique (model, country, period,* θ*) task cells yields* $\Delta = +2.10$ *pp (*$t = 6.383$*,* $p < 0.001$*).*[3] *Equal-weighted model averages are close to zero under both summaries (*-0.3 *pp unpaired;* -0.09 *pp paired), reflecting heterogeneous model-specific effects; point estimates are negative in three of seven models.*

The pooled estimate is dominated by the primary model; on the equal-weighted model average the communication effect is approximately zero. Communication preserves the signal structure ($r = +0.78$, against $+0.80$ under the pure treatment; see Table 2, and Figure 4 for the effect by regime strength). Simple text features in baseline messages predict θ, so peer messages carry real, if noisy, information. The mean communication effect is modest because messages can transmit both encouragement and caution depending on the signal realization, which is also why the equal-weighted model average sits close to zero. This modest average effect is not the surveillance estimand. The surveillance question, to which I turn in the next section, is whether changing the sender-side rule changes the downstream message environment enough to move actions.

A *degraded-messages* control replaces all LLM-generated messages with generic uninformative text while preserving the two-stage protocol and omitting any surveillance framing. Against its own live-messages baseline ($N = 500$, mean join 45.9%), degraded messages drop the mean join rate to 24.5% ($\Delta = -21.4$ pp, $p < 0.001$), well below even the pure game (38.7%). This is consistent with communication quality mattering, but the generic text may itself convey caution. A no-message control therefore provides a cleaner zero-peer-message benchmark against the main communication baseline, with join rates of 37.1% ($\Delta = -4.0$ pp; $N = 500$).

6 Surveillance

I now turn to the central question of the paper: whether an authoritarian regime can suppress coordinated action by degrading the communication channel alone. In the clean

[3] "Matched-cell" and "paired" refer to the same estimator throughout: treatment arms are compared on cells with identical task draws. The exact matching key also includes the signal draw and payoff context; see the notes to Table A7.

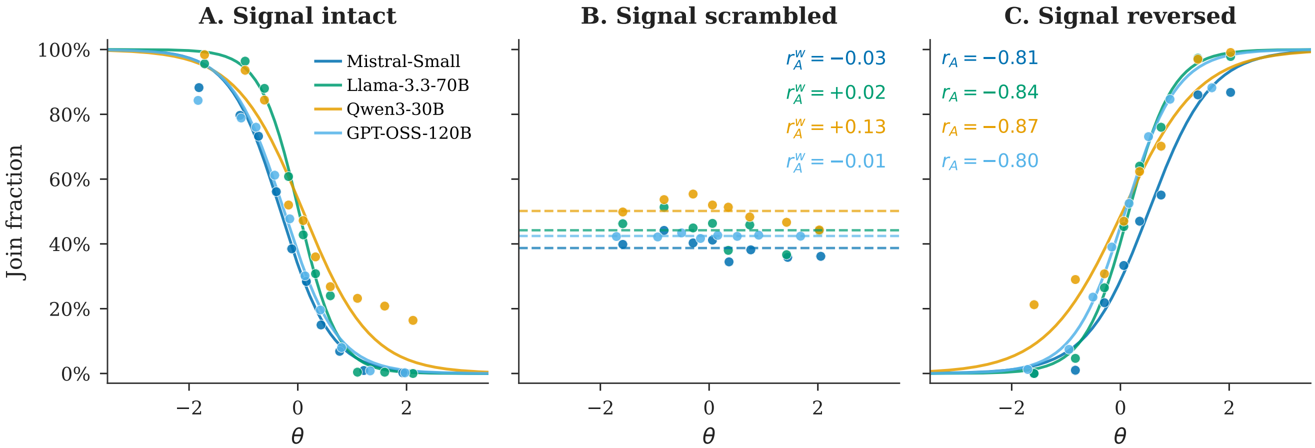

Figure 3: Falsification triptych. Plotted series show four representative model families; reported mean correlations are across all 7 models in Table 2, with per-arm period counts in Table A19. *Left:* Pure (mean $r = +0.80$, raw per-model). *Center:* Scramble ($r = +0.03$, within-country). *Right:* Flip ($r = -0.80$).

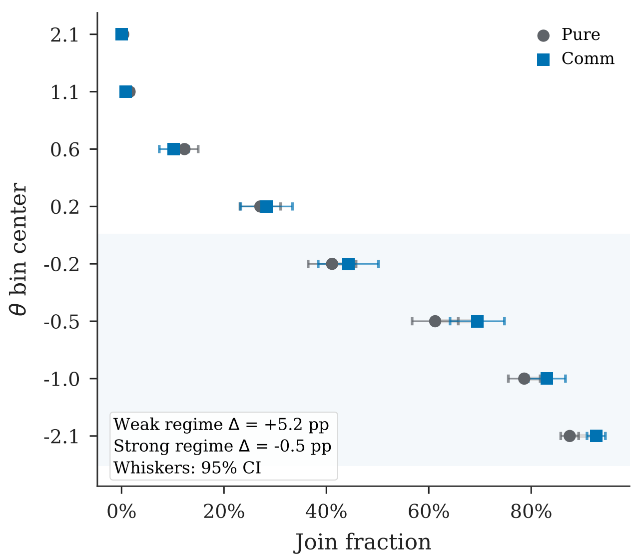

Figure 4: Communication effect by regime strength, pooled across 7 models. Points are mean join rates by θ bin under the pure and communication treatments; whiskers are 95% confidence intervals on bin means; the shaded region marks weak-regime states ($\theta < 0$). In-figure Δ annotations are in percentage points.

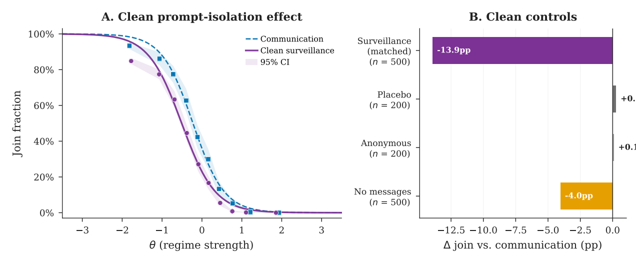

Figure 5: Clean primary-model surveillance and message controls (Mistral). *Panel A:* Join curves under communication baseline vs. clean surveillance; shaded bands are 95% confidence intervals; surveillance shifts the curve downward while preserving threshold structure. *Panel B:* The clean-surveillance bar uses the matched-cell estimator ($n = 500$ matched cells); placebo ($n = 200$), anonymous ($n = 200$), and no-message ($n = 500$) bars are control deltas against the main communication baseline; per-bar sample sizes appear on the axis labels, and inference is reported in Tables A9 and A10. Placebo and anonymous monitoring warnings sit near zero.

surveillance treatment (equivalently, the "clean prompt-isolation rerun"), the communication prompt is the baseline trusted-contact prompt plus one appended warning that communications are being monitored by regime security services. The manipulation affects only the communication phase; the decision prompt is unchanged. The isolation is architectural, in that each LLM call is stateless: the decision-stage API call receives only the briefing and the messages, with no memory of the communication-stage surveillance warning. The treatment therefore asks whether changing the *sender-side* message rule is enough to change receiver actions.

Result 5 (Surveillance Reduces Participation)**.** *The clean prompt-isolation rerun reduces mean join rates in the primary model (Mistral Small Creative) from 45.3% to 31.4%, a shift of* -13.9 *pp (*$p < 0.001$*;* $N = 500$*). The action–attack-mass correlation is preserved (*$r = +0.72$ *vs.* $r = +0.74$ *under the communication baseline), indicating that surveillance operates as a level shift rather than disrupting signal processing.*

Surveillance changes the information environment by replacing trusted-contact messages with monitored-contact messages, pushing join rates below the communication baseline (Figure 5). The clean primary-model matched-cell estimate is a 13.9 pp decline; pooling matched cells across all five rerun models gives $\Delta = -13.5$ pp ($N = 580$, $p < 0.001$). Placebo variants using the same baseline-plus-warning template—monitored for research (no consequences) and aggregated anonymously—produce no significant deviation from the communication baseline in the primary model (+0.3 pp and +0.1 pp, both $p > 0.9$; Appendix E.4); Llama 3.3 70B placebo variants are small but imprecisely estimated. Across the clean prompt-isolation rerun set, 5/5 models move in the negative direction on matched-cell support (equal-weighted mean $\Delta = -11.2$ pp; 4/5 individually significant at 5%; Appendix Table A9). The non-primary rows are sign checks on only 20 matched cells each (the original grids were not nested); Qwen3 235B is smallest in magnitude and individually insignificant ($\Delta = -3.0$ pp, $p = 0.186$). The full-scale evidence is the primary-model result and the nested 500-cell Llama replication below.

6.1 A Nested Replication and a Codedness Control

Because the original cross-model reruns shared only 20 cells with their baselines, I ran a fully nested replication campaign on Llama 3.3 70B, an endpoint that remains publicly accessible. The campaign centers on a communication baseline, a clean surveillance arm, and a codedness-induction control on an identical 500-cell grid (10 countries $\times$ 50 periods, $n = 25$, common seed), so that every cell is matched by construction. Surveillance reduces joining from 47.6% to 39.6% ($\Delta = -8.0$ pp, $t = -16.24$, $p < 0.001$, $N = 500$ matched cells), a full-scale, reproducible confirmation of the primary-model result on a second model family. Graded re-phrasings of the warning on the same grid show that the effect does not depend on the particular warning sentence: joining falls monotonically in the stated severity of monitoring, and most of the decline already arrives under a mild routine-monitoring note with no consequence language (Appendix E.5).

The codedness control instructs senders to write in indirect, metaphorical language, with no mention of monitoring or consequences. The instruction binds: 97.0% of control messages contain coded-metaphor terms and only 0.8% contain direct regime references, making them *more* coded on these dictionaries than the surveilled messages themselves (25.3% coded, 68.5% direct). Yet joining barely moves ($\Delta = -0.7$ pp, $p = 0.007$; a small but precisely estimated shift), and the gap between the surveillance arm and the codedness control on the same cells is -7.4 pp ($p < 0.001$). Coded form per se is therefore not what suppresses participation, and a generic-vagueness reading of the surveillance effect is ruled out.

A decoding test asks the reverse question: if the coded form is stripped away, how much participation returns? A different-family model (Qwen3 30B) rewrites each surveilled message in plain, direct language, instructed to preserve the factual content, and both the raw and the decoded messages are replayed to fresh receivers on the same cells through identical fixed-message machinery. The raw replay reproduces the live arm ($\Delta = -7.9$ pp, against -8.0 pp live), validating the replay design; decoding restores +3.1 pp of participation relative to the raw messages ($p < 0.001$; $\Delta = -4.9$ pp against baseline). Together the two manipulations bracket the mechanism. Coded form alone suppresses almost nothing; stripping the coded form from surveilled messages restores roughly two-fifths of the effect; and the remainder travels with content that survives translation—the guarded, hedged substance of what monitored senders choose to say. Surveillance damages both what is said and how legibly it is said.

Interpreting this result requires separating four possible channels. First, the warning could deter receivers directly, but the stateless decision prompt rules this out in the main treatment; the hold-fixed test below measures direct deterrence only when the warning is deliberately moved to the decision stage. Second, monitoring could remove information about regime weakness, which the content evidence tests by asking whether bad news disappears or merely changes form. Third, it could degrade messages for downstream judgments generally; the private-bet discriminant test shows movement on a non-action decision label, although a bet on regime fall can still depend on beliefs about others' participation and so does not isolate a structurally non-coordination component. Fourth, it could reduce the public, action-cueing value of messages—the higher-order channel targeted by the belief-elicitation contrast, the no-message controls, the coded-language shift, and the cross-model writer–reader rotation. Appendix Table A3 summarizes the full map.

6.2 Locating the Channel: Communication Poisoning

By "communication poisoning" I mean something specific: monitored messages become less useful inputs for the receiver's coordination decision. In global-game terms, the treatment can leave information about the fundamental θ in the message while changing what the message says about strategic complementarity: whether others are willing to condition on the same signal. In LLM terms, the prompt intervention changes the conditional distribution of generated peer messages, while the receiver prompt and private briefing are held fixed. Three distinct message properties could change: (a) factual information about regime strength θ, (b) coordination cues about what other citizens are willing to say or do, and (c) risk/fear salience transmitted by tone. The evidence is not that surveillance simply deletes bad news about the regime. Monitoring changes the language of peer messages, and the changed messages are less action-guiding for receivers; the dictionary and classifier evidence below identifies *what* changed in the text, not which textual property drives the action response. The codedness control above shows that metaphorical form alone does not suppress participation, and the receiver-rationale audit weighs against fear contagion; the evidence points instead to a loss of mutual legibility, in which surveilled messages stop signaling that others are willing to act. What remains bundled is which specific property of the surveilled text carries that signal (Table A3).

Message content changes systematically, and in one direction. A logistic classifier on uni- and bi-grams distinguishes baseline from surveillance messages with 91.9% accuracy on a held-out sample of $N = 15{,}000$ messages (chance = 50%; primary model). The same finding replicates on class-balanced Llama 3.3 70B and Qwen3 30B samples (98.5% and 93.6% accuracy, respectively); in 3/3 measured models, direct references fall and coded-metaphor references rise. The shift is not toward fewer explicit weakness references: in the primary model, references to regime weakness *rise* from 23.2% to 31.7% of messages ($z = 21.06$), while strength references drop from 14.5% to 8.5%. Rather, the shift is from direct to coded language: direct references to the regime, security forces, and streets fall from 63.9% to 34.5% of messages ($\Delta = -29.4$ pp), replaced by metaphorical phrasings ("walls are cracking," "ground shifting," "heads down," "feels different"; 69.8% to 94.1%, $\Delta = +24.3$ pp). The n-gram classifier in Appendix Table A6 shows separability of the peer-message distributions; the direct-reference and coded-metaphor rates are separate dictionary counts. Within the surveillance arm, codedness is the measured textual signature of the treated message distribution, with weakness cues present but less direct; the codedness control above randomizes the coded form itself. With the decision prompt unchanged, the behavioral effect is localized to this sender-generated message distribution.

A further interpretive caveat applies on the sender side. The response to the monitoring warning may partly reflect alignment training—caution about producing subversive content when monitoring is salient—rather than the strategic self-censorship of a simulated citizen. Trinity Large's elevated provider content-filter failures (Table A19) are direct evidence that safety layers activate on this material. The design identifies the downstream effect of the changed messages on receivers, not the sender's motive for changing them.

A dictionary audit of receivers' free-text decision rationales sharpens the receiver-side picture. Explicit references to monitoring or surveillance are rare in both arms (1.0% of 24,998 valid baseline rationales vs. 1.5% of 25,000 under surveillance), so the action response does not run through receivers verbalizing an inference that they are watched. Nor do rationales become more fearful: fear and caution references *fall* from 84.2% to 69.8% ($p < 0.001$), weighing against fear contagion through message tone as the dominant receiver-side channel. What changes is engagement with the peer channel itself: references to contacts and peer messages drop from 28.5% to 15.8% ($p < 0.001$), while regime-weakness references rise from 85.3% to 91.2%. Receivers still read bad news but lean less on the messages when those messages arrive coded—the pattern the actionability interpretation predicts. These are post hoc dictionary counts on stated rationales, which need not reflect the model's actual computation.

The degraded-messages control (Section 5, Communication) exogenously replaces all messages with generic text, without any surveillance framing, and drops joining by 21.4 pp (from its 45.9% live-messages baseline to 24.5%). That comparison uses its own live-message rerun. Two implications follow. First, decision-stage agents use message content. Second, replacing real messages with generic text depresses participation on its own—the degraded-messages drop exceeds the surveillance effect itself—which is why the codedness control matters: induced metaphorical style without monitoring framing leaves joining nearly unchanged, so the surveillance effect is not generic message degradation. The no-message control uses the main communication baseline and gives a cleaner zero-peer-message benchmark (primary model: 37.1%; $\Delta = -4.0$ pp, $p = 0.0467$). Across the three models with no-message data (Mistral, Llama 3.3 70B, Qwen3 30B), removing peer messages reduces joining in all three cases (-4 to -16 pp), though the primary-model estimate is the smallest and only marginally significant; this is consistent with decision-stage agents weighting peer messages as informational input.

Where surveillance lies relative to this floor varies. In Mistral it falls below silence; in Llama and Qwen3 30B it remains above silence but below baseline communication. Across models, then, monitored messages are not always worse than silence, but monitoring degraded the communication channel in all three models examined; the below-silence pattern, which suggests active poisoning beyond information removal, appears only in the primary model.

Elicited punishment risk is $\approx 8.0/10$ across all conditions with negligible differences between JOIN and STAY agents (Table A16). Because the elicitation sits near the top of the scale in every condition, the instrument may be ceiling-limited; it does not separately identify fear salience in message tone, but it weighs against a simple receiver-side story in which surveilled messages mainly work by making agents report higher punishment risk.

6.3 Belief Elicitations and Actions

I elicit second-order beliefs (SOB) with the question, "What percentage of citizens will choose to JOIN?" The *primary belief sample* is the post-decision elicitation in which the agent does not re-see the peer messages on the belief prompt; this keeps the instrument from mechanically re-exposing the treated text after the action. In that sample, communication agents report 31.2% and surveillance agents report 30.9% ($\Delta = -0.3$ pp, $p = 0.5866$), while joining falls from 41.4% to 28.5% ($\Delta = -12.9$ pp).

Stability here is a bounded contrast, not only a failure to reject: the agent-level 95% confidence interval on the SOB shift, $[-1.3, +0.7]$ pp, confines it to about one percentage point, an order of magnitude smaller than the 12.9 pp action shift. First-order success beliefs are also close (44.4% vs. 43.6%). A point report of expected participation is not the whole global-game object: action value also depends on confidence that other receivers parse the same messages as action-relevant, and that object can move when a coarse report of expected joining does not. The same agents who report nearly unchanged expectations about others' participation choose substantially less participation themselves.

The belief evidence is sensitive to the elicitation information set. When the belief prompt includes the received peer messages, reported beliefs and SOB move sharply under surveillance (Appendix C, Table A2); a parallel pre-decision elicitation in the same messages-included condition produces $\Delta\,\mathrm{SOB} = -8.9$ pp ($p < 0.001$), close to the post-decision contrast (-8.2 pp). The pre-decision $\times$ messages-excluded cell could not be collected on the primary model before its endpoint was retired (Appendix F), but the nested Llama 3.3 70B campaign collects exactly this cell: with pre-decision elicitation and messages excluded, first-order beliefs are 41.7 vs. 41.9 ($p = 0.550$) and second-order beliefs are 30.2 vs. 30.2 ($p = 0.983$), while joining falls 8.0 pp on the same cells. The belief/action wedge therefore survives a pre-decision elicitation in the restricted information set and is not a post-decision rationalization artifact. For the primary model itself, timing effects cannot be ruled out from the archived cells alone, though simple post-decision rationalization would tend to push expected participation downward when surveillance lowers joining and so does not explain a near-zero messages-excluded contrast; the Llama pre-decision result indicates the wedge is not an artifact of elicitation timing.

A separate hold-fixed test replays baseline messages verbatim while adding a surveillance warning to the *decision* prompt, shifting joining by -20.9 pp ($p < 0.001$). This shows a direct decision-stage channel exists, but it is absent in the main design, where the decision prompt contains no surveillance reference. The main treatment isolates the indirect channel: communication poisoning.

6.4 A Task Contrast: Private Bets versus Coordination Decisions

A complementary discriminant test holds the surveillance manipulation fixed but swaps the task. In Llama 3.3 70B, I keep the briefing pipeline and messaging stage identical, including the surveillance prompt, but reframe the decision-stage prompt as a private bet on regime fall, whose payoff is independent of the respondent's own action. Surveillance still moves the bet by -5.6 pp ($p = 0.0256$; $N = 500$ per condition; baseline 34.7% vs. surveillance 29.1%), so the treated messages affect downstream judgment even when the respondent is not choosing the strategic action. The full Llama prompt-isolation coordination-task estimate is larger (-15.0 pp), but the samples are not cell-matched and a bet on regime fall can still embed participation expectations, so I read this as a descriptive task contrast rather than a decomposition.

6.5 Cross-Model Writer–Reader Rotation

A standing concern with the within-model design is reflexivity: the writer model and the reader model are the same, so the reader could be responding to stylistic artifacts of its own family rather than to a content property of surveilled messages. I therefore test bidirectional rotation on matched (country, period, θ) cells. *Llama writes, Qwen3 30B reads*: surveillance moves Qwen joining by -7.5 pp ($p < 0.001$, $N = 200$), comparable to the within-Llama matched effect (-6.0 pp on identical cells). *Qwen3 30B writes, Llama reads*: the reverse direction gives a smaller but still significant cross-model effect (-1.9 pp, $p < 0.001$, $N = 200$). Both directions rule out the reader merely recognizing its own family's style: at least part of the effect transmits across model families through message content alone. The cross-direction asymmetry is unexplained, and its magnitude cautions against treating the full within-model effect as content-driven.

7 Conclusion

How surveillance suppresses dissent is a central question in the political economy of authoritarian rule, and it has a game-theoretic core: collective action requires not only information but coordination on what that information means. This paper has presented an experiment at that intersection, a controlled global game in which monitoring can suppress coordinated action without ever appearing in the receiver's decision problem. LLM citizens first reproduce the canonical comparative static, with participation falling as regime strength rises, and communication then turns private prose signals into social information.

When monitoring enters only during that messaging stage, joining falls by 13.9 pp in the primary model and by 8.0 pp in a fully nested 500-cell replication on Llama 3.3 70B, with negative point estimates in all five clean prompt-isolation reruns, even though reported expectations about how many others will join remain approximately stable, both post-decision in the primary model and pre-decision in the nested replication (peer messages are not re-shown in either elicitation). Placebo warnings do not replicate the effect in the primary model, and message controls show that downstream agents use peer messages as part of their decision environment. The same channel that transmits coordination-relevant information is thus also the channel an authoritarian regime can degrade, by damaging the conversion of information into a shared reason to act rather than by removing the information itself.

The mechanism evidence extends beyond the main treatment contrast. Surveillance moves a private-bet task as well, so the treated messages affect downstream judgment outside the JOIN/STAY action prompt, though that comparison is descriptive. Message text also shifts from direct statements toward metaphorical and deniable phrasing, and the codedness-induction and decoding tests bracket what that shift does: more heavily coded messages produced without monitoring framing leave joining nearly unchanged, while translating surveilled messages back into direct language restores roughly two-fifths of the lost participation, with the remainder traveling with the guarded substance of what monitored senders choose to say. In language-mediated global games, then, surveillance can degrade the higher-order content of information without fully degrading its first-order content: it can leave bad news in circulation while making that news less mutually legible as something others will act on. More broadly, the experiment isolates a channel by which repression could target the publicness and actionability of information, not just its factual content. Whether this characterizes how authoritarian regimes *actually* deploy monitoring is a separate empirical question; the contribution here is to isolate the communication-poisoning channel under controlled prompts and matched decision contexts.

Four limitations remain, and they suggest avenues for further research. First, training-data familiarity (Section 4): whether the behavior reflects approximate Bayesian reasoning or learned patterns remains open, and the sender-side response may partly reflect alignment training rather than simulated self-censorship. Second, although graded re-phrasings of the warning produce a monotone dose response with most of the effect already present at a mild routine-monitoring note (Appendix E.5), all variants share the same appended-note format, so sensitivity to structurally different framings of monitoring remains untested. Third, the experiments are static (Angeletos et al., 2007; Chassang, 2010); in repeated interaction receivers may learn to decode euphemism (and senders to re-code), which would make the static effect an upper bound. Fourth, and deepest, is external validity: what LLM agents do in this game is evidence about how language-trained agents respond to the information structure, not necessarily about how humans would play it, and the primary model's retired endpoint means its arm can be re-analyzed from the archived cache but not re-queried. The nested 500-cell Llama replication and the cross-model benchmark show that the pattern is not specific to one model family. The experimental template is reusable, and the data and code are provided in the de-identified replication archive.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the author used OpenAI Codex to assist with code review, consistency checks, replication-package auditing, and manuscript editing. The author reviewed and edited the output and takes full responsibility for the content of the published article. This statement concerns manuscript preparation only; the LLM calls analyzed as experimental data are described in the methods and archived in the replication materials.

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A Experimental Design Details

Table A1 reports the five-hypothesis family and outcomes. Results 1–5 in the main text correspond to H1 (alignment), H2 (scramble falsification), H3 (flip), H4 (communication), and H5 (surveillance); the four directional results (H1, H3–H5) survive Bonferroni correction at $\alpha = 0.01$.

Narrative header texts. The eight narrative stakes-ladder conditions prepend the following headers to each agent's standard intelligence briefing (the briefing body is identical across conditions):

> **Very high cost:** "CONTEXT: The regime maintains an extensive network of informants and has a documented history of disappearances following failed uprisings. Participants' families are routinely targeted. The cost of unsuccessful action is existential."

> **Moderate high cost:** "CONTEXT: Participants in past failed uprisings have faced prison sentences of several years and loss of employment. The personal cost of unsuccessful action is significant but not life-threatening."

> **Neutral:** "CONTEXT: The consequences of participating in a failed uprising are uncertain and vary case by case. Some participants have faced penalties while others have not. The personal stakes are ambiguous."

> **Moderate low cost:** "CONTEXT: Recent legal reforms have limited the regime's ability to punish dissent. Past participants faced only brief questioning and were released without charges. The personal risk of action is modest."

> **Very low cost:** "CONTEXT: The international community has guaranteed safe passage and asylum for all participants regardless of outcome. Even failed uprisings carry essentially zero personal cost."

> **Placebo:** "CONTEXT: The country has experienced significant weather disruptions this year, including severe flooding in the eastern provinces and an extended drought in agricultural regions. Infrastructure damage has been substantial."

Paraphrase variants (high cost v2, low cost v2) use different surface wording to express the same cost level.

B Complete Prompt Example

The following shows the complete text that an LLM receives for a single decision. The *decision prompt* is:

> "You are a citizen deciding whether to JOIN an uprising or STAY home. You only have your private briefing below. You do not know how many others will JOIN. Start your first line with exactly JOIN or STAY, then give 1–2 concise sentences of reasoning."

This is followed by the agent's intelligence briefing (see Figure A9 for complete examples at $z \in \{-2, 0, +2\}$). In the *communication* treatment, agents first receive:

> "You are a citizen in a country with an authoritarian regime. You have received a private intelligence briefing. You are now in contact with a small number of trusted people. Based on your briefing, write a brief message (1–2 sentences) to share your read of the situation. Be natural—you're talking to people you trust."

After exchanging messages on the network, agents receive a separate communication-decision prompt:

> "You are a citizen deciding whether to JOIN an uprising or STAY home. Use your private briefing and messages from trusted contacts. You do not know how many others will JOIN. Start your first line with exactly JOIN or STAY, then give 1–2 concise sentences of reasoning."

The user message then contains the agent's briefing under "YOUR INTELLIGENCE BRIEFING" and the neighbor messages under "MESSAGES FROM TRUSTED CONTACTS." The pure-only sentence "You only have your private briefing below" does not appear in communication decision calls. In the *surveillance* variant, the communication prompt appends:

> "IMPORTANT: You have reason to believe that your communications are being monitored by regime security services, and messages deemed subversive could have serious consequences for you and your contacts."

The decision prompt is *unchanged* under surveillance. It contains no mention of monitoring. Any behavioral effect therefore operates through the content of the (now self-censored) messages.

C Belief Elicitation

Beliefs are elicited on a 0–100 probability scale immediately after each decision. The first-order question is "How likely is the uprising to succeed?" The second-order question is "What percentage of citizens will choose to JOIN?" The primary belief elicitation prompt includes the private briefing and the agent's own decision, but not the received peer messages; this keeps the elicitation from mechanically re-exposing the treatment text after the action. The main-text values (31.2% vs. 30.9%; join 41.4% vs. 28.5%) come from this original primary belief run and Figure A2. Table A2 reports a separate 500-cell factorial rerun designed to test information-set sensitivity;

Table A1: Hypothesis family and test results. H1–H4 use pooled benchmark-phase data across all seven models; H5 uses the primary model (Mistral Small Creative). Effect size: r for correlations (H1–H3), Cohen's d or d_z for mean comparisons (H4–H5). Outcome labels use $\alpha = 0.05$; H2 is reported as a falsification check, not as evidence for a zero effect.

H	Hypothesis	Test	Statistic	N	p	Effect	Outcome
H1	Sigmoid Response	Pearson	0.757	1,800	<0.001	0.757	Supported
H2	Scramble Falsification	Pearson	0.014	1,200	0.638	0.014	Passes falsification
H3	Directional Sensitivity	Pearson (1-sided)	−0.791	1,100	<0.001	−0.791	Supported
H4	Communication Channel	Paired t	6.383	1,479	<0.001	0.166	Supported
H5	Surveillance Chilling	Paired t	−19.109	500	<0.001	−0.855	Supported

Notes: The Test column gives the statistic type for each row: Pearson r for H1–H3 (the Statistic column reports r itself) and paired t-tests for H4–H5 (the Statistic column reports t). N counts period-level rows for H1–H3 (H2 uses the pooled scramble arm) and matched task cells for the paired tests H4–H5. H1–H4: pooled benchmark-phase data across all seven models (H4 uses a paired t-test matched on model/country/period/θ task cells). H5: primary model (Mistral Small Creative), paired on matched prompt-isolation cells. Bonferroni survivors at $\alpha = 0.01$: H1, H3, H4, H5.

its messages-excluded cell has the same qualitative belief/action contrast but different sample means. Figure A1 documents belief alignment with the theoretical posterior ($r = +0.80$); Figure A3 confirms monotone signal–belief mapping nonparametrically.

A factorial variant tests information-set sensitivity. Table A2 crosses condition (communication vs. surveillance) with elicitation information set (messages included vs. excluded). Without messages, first-order beliefs barely shift under surveillance (−0.3 pp, $p = 0.34$) and second-order beliefs are effectively unchanged (0.0 pp, $p = 0.89$), while actions shift −11.3 pp. With messages included, beliefs shift significantly (−5.8 pp and −8.2 pp, both $p < 0.001$). Surveillance does not leave all beliefs unchanged under every elicitation prompt; the wedge is a property of the restricted information set, where it holds both post-decision (here) and pre-decision (nested replication, Section 6). Figure A2 shows second-order beliefs under the message-excluded information set.

D Identification Test Details

Table A3 summarizes which candidate channels each test addresses and what the test concludes. The remaining tables in this section identify which text surface changes: briefing-body classifiers test whether the unchanged private briefing text can account for the action shift, while the peer-message classifier supports the message-stage degradation row.

Contamination and demand audit. Following Horton (2023), I probe directly whether models recognize the experiment. For 4 accessible paper models, 10 fresh post hoc calls per model per question present the decision prompt and a $z = 0$ briefing without asking the model to play. Asked whether the setup resembles a formal model, 5% of responses name the global game or Morris–Shin specifically, while 78% identify a coordination or collective-action setting generically. Asked to predict the effect of the surveillance manipulation when it is described explicitly, 82% of responses predict lower participation. Hypothesis recognition is therefore available to the models when prompted, and the design does not rely on participants being unaware. The receiver-side estimate is instead identified by the architectural exclusion of the warning from the decision prompt, and the sender-side variants show the response tracks the attribution of monitoring to the authorities rather than the mere mention of monitoring: research-framed and anonymized monitoring leave joining essentially unchanged, while even a consequence-free routine-monitoring note attributed to the authorities reproduces most of the effect (Appendix E.5). The audit cannot be run on the retired primary model.

Table A4 reports an action-prediction diagnostic, not a baseline-versus-surveillance condition classifier. Classifiers trained on pure-treatment briefing-body text predict how agents would act from the private briefing alone; when applied to surveillance-treatment briefings, they predict join rates near the non-surveilled benchmark and miss the observed action drop. Table A5 asks a parallel question for narrative stakes headers: whether baseline briefing-body features can predict header-induced behavioral shifts. They cannot, which is expected because the headers are outside the briefing-body feature set. In contrast, a classifier on the actual peer messages, which are LLM-generated and downstream of the surveillance prompt, separates the two conditions sharply (91.9% accuracy on $N = 15{,}000$ held-out messages; Table A6). Together the classifier exercises locate the manipulated text surface: private briefing text does not explain the action shift, stakes headers change behavior outside the baseline feature set, and surveilled peer messages change visibly.

E Robustness

Threshold-policy alignment remains strong across agent counts ($r \in [+0.67, +0.73]$ across $n \in \{5, 10, 50, 100\}$, somewhat below the main-run range), network density ($r = +0.69$ at $k = 8$ vs. $+0.74$ at $k = 4$), and mixed-model populations ($r = +0.86$ pure, $r = +0.83$ communication); Figure A4 plots the first two checks.

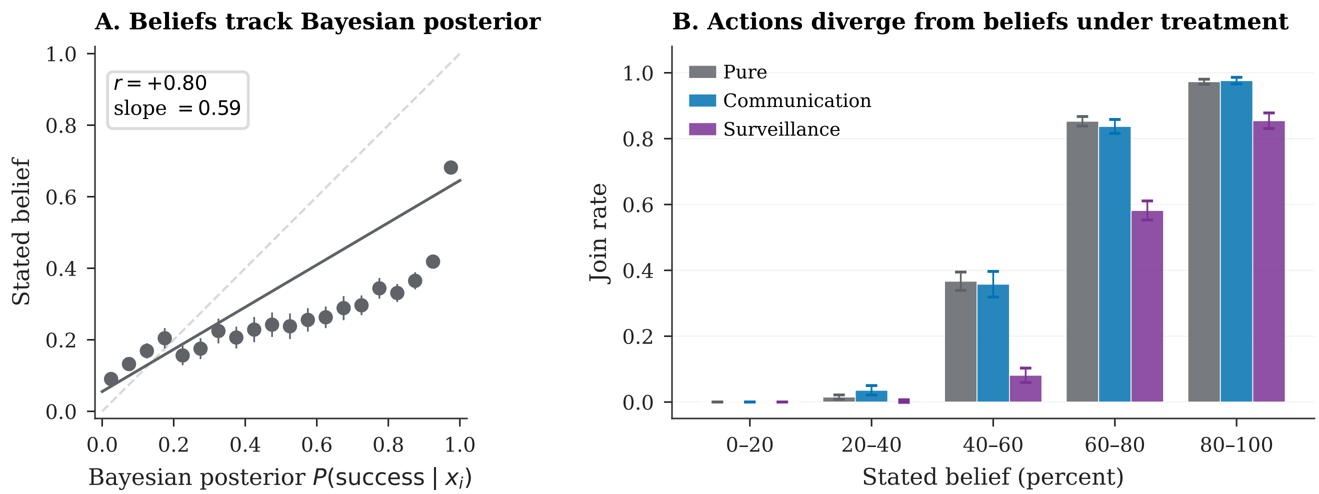


Figure A1: First-order belief elicitation (Mistral, pure treatment, $N = 10{,}000$ agent-level observations). *Left:* Stated beliefs vs. theoretical $P(\text{success} \mid x_i)$, binned means with 95% confidence intervals ($r = +0.80$). *Right:* Join rate by belief bin. Second-order belief/action contrasts are shown in Figure A2 and Table A2.

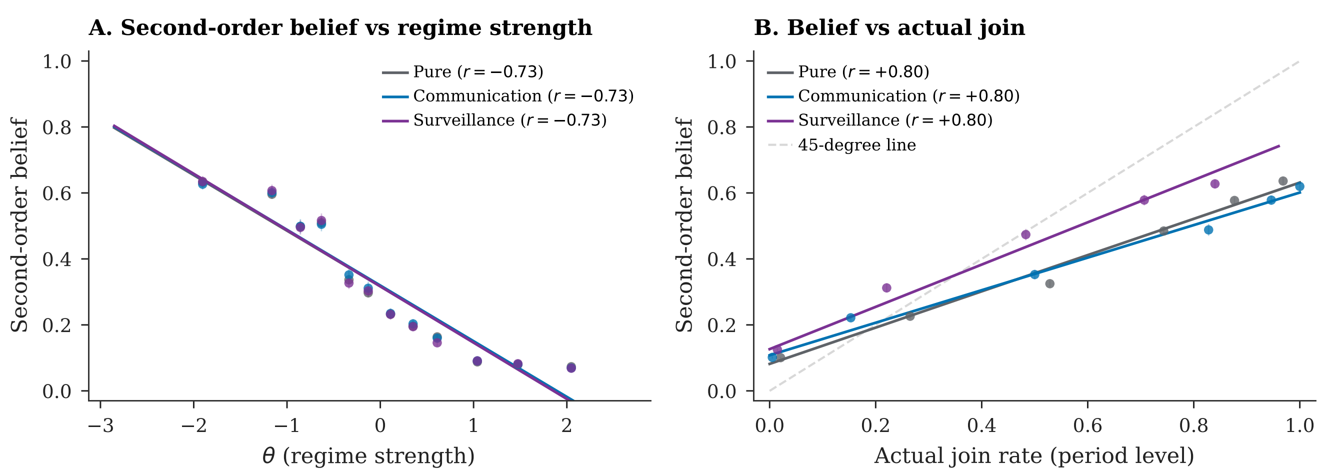


Figure A2: Second-order beliefs (Mistral, original primary belief run; restricted information set, messages excluded; binned means with 95% confidence intervals). *Panel A:* plotted against regime strength, the treatment arms largely overlap. *Panel B:* plotted against actual join rates, the surveillance curve sits higher because actions fall while restricted-information second-order reports move little. The factorial in Table A2 shows a significant shift when messages are included.

Table A2: Belief/action contrast under surveillance, primary model (Mistral Small Creative): effect of including peer messages in the belief elicitation prompt ($N = 12,500$ agent-level observations per cell).

	Messages excluded		Messages included	
	Comm	Surv	Comm	Surv
First-order belief	46.0	45.7	53.2	47.4
Second-order belief	32.7	32.7	42.0	33.8
Join rate	42.4%	31.0%	42.2%	30.9%
Surveillance effect (comm $\to$ surv):				
Δ belief	-0.3 ($p = 0.34$)		-5.8 ($p < 0.001$)	
Δ SOB	0.0 ($p = 0.89$)		-8.2 ($p < 0.001$)	
Δ join	-11.4 pp ($p < 0.001$)		-11.3 pp ($p < 0.001$)	

Notes: Post-decision elicitation, 500 country–periods × 25 agents per cell ($N = 12,500$). "Messages excluded": belief prompt sees briefing and decision only. "Messages included": belief prompt additionally sees the peer messages received in the communication round. Beliefs and SOB are on the 0–100 scale; join rates in percent. p-values are from two-sample Welch t-tests on agent-level observations (unclustered). Deltas are differences of the displayed cell means.

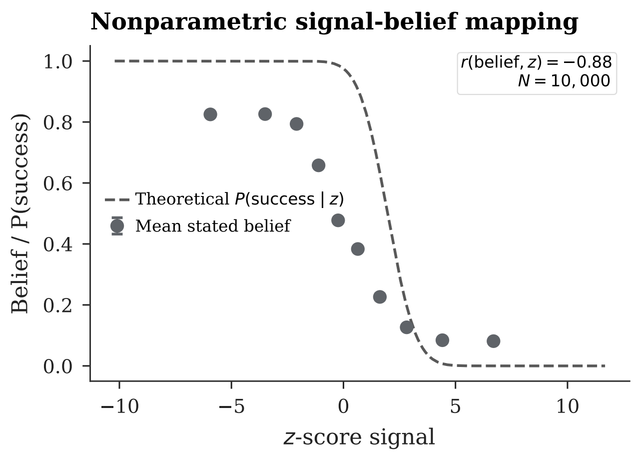
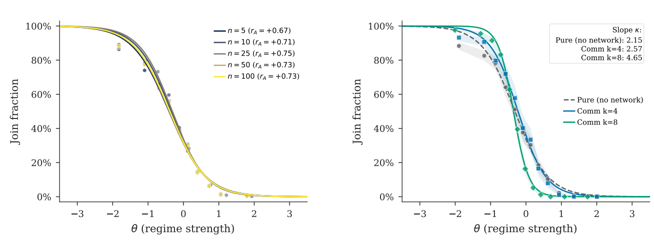


Figure A3: Nonparametric monotonicity (Mistral, pure treatment): mean stated belief by z-score bin vs. theoretical benchmark, with 95% confidence intervals on bin means. The in-figure correlation $r(\text{belief}, z)$ is the belief–signal correlation, a different object from the belief–posterior correlation $r = +0.80$ reported in the text.



(a) Agent count ($n \in \{5, 10, 50, 100\}$).

(b) Network density ($k = 4$ vs. $k = 8$).

Figure A4: Robustness checks for threshold-policy alignment (Mistral). Points are period-level join fractions with fitted logistic curves. In panel (a), legend entries report $r(J, A(\theta))$ per agent count; in panel (b), the inset reports fitted logistic slopes κ and shaded bands are 95% confidence intervals.

E.1 Communication Aggregation

Communication aggregation accounting. Table A7 decomposes the communication estimators by model. The matched-cell estimand is the main one because it compares identical task cells. It keeps essentially all pure-treatment cells but drops communication-only cells in the larger communication arms; the gap relative to the pooled unpaired contrast is therefore driven mainly by repeated-cell weighting and arm-size differences, especially in the largest model, rather than by pure-treatment support loss.

E.2 Generator Robustness

Replacing the logistic function mapping z-scores to directional sentiment with three alternatives—tanh, piecewise linear, and a hard step function at $z = 0$—again yields high attack-mass correlations: $r = +0.83$ (tanh), $r = +0.83$ (linear), $r = +0.84$ (step) on $N = 100$ country–periods each, compared with $r = +0.75$ for the logistic baseline (main pure sample, $N = 1000$). A domain ablation test removes subsets of the briefing generator's eight evidence domains. Removing the two coordination-language domains raises mean joining by +3.2 pp while preserving the threshold relationship ($r = +0.90$). Removing the four regime-state domains lowers mean joining by 3.0 pp and attenuates the threshold relationship ($r = +0.86$). These ablations show that threshold-like signal use is not knife-edge in the coordination-language domains, but they are not surveillance-channel ablations: the clean communication and surveillance reruns still use the full briefing generator, including atmosphere and actionability cues. A stronger isolation test would rerun baseline communication and surveillance while holding those coordination cues fixed or removed.

E.3 Decoding Temperature

The correlation $r(J, A(\theta))$ ranges from +0.78 to +0.86 across 13 model–temperature combinations (up to five temperatures per model, $T \subseteq \{0.3, 0.5, 0.7, 1.0, 1.2\}$; Table A8).

E.4 Falsification and Placebo Tests

Two clean placebo variants—monitoring "for research purposes" (no consequences) and "aggregated anonymously"—produce no significant deviation from the communication baseline in the primary model (+0.3 pp, $p = 0.9351$ and +0.1 pp, $p = 0.9790$; Table A10). Llama variants are small but imprecise. The large decline is therefore tied to monitoring attributed to the regime rather than to generic observation; the dose-response check below shows that explicit consequence language is not necessary for most of the effect. As a conservative inference check on the primary contrast, averaging the 500 matched-cell deltas within each country and testing across the 5 country means present in the matched sample gives $t = -13.90$ ($p < 0.001$), so the headline effect survives clustering at the country level. The effect is also concentrated where coordination is pivotal: splitting the 500 matched cells by realized-θ tercile, the decline is -19.5 pp in the weak-regime tercile ($\theta \in [-2.96, -0.46]$, $p < 0.001$) and -20.3 pp in the middle tercile ($\theta \in [-0.46, +0.44]$, $p < 0.001$), but an order of magnitude smaller—though still detectable—in the strong-regime tercile (-2.1 pp, $p < 0.001$), where few agents join under either arm. This is the pattern expected if surveillance degrades coordination cues rather than shifting behavior uniformly.

E.5 Warning Phrasing and Dose Response

The clean surveillance treatment uses a single warning phrasing, so a natural concern is that the effect is tied to that sentence. Two graded variants rerun the nested Llama 3.3 70B campaign on the identical 500-cell grid, changing only the appended warning. The *mild* variant reads "NOTE: Communications in your area are sometimes subject to routine monitoring by the authorities," removing the consequence language entirely; the *severe* variant reads "WARNING: Your communications ARE being monitored by regime security services. Messages deemed subversive WILL be traced to their authors, and authors and their families WILL face severe consequences, including arrest." Joining falls by 6.8 pp under the mild warning ($t = -13.38$, $p < 0.001$, $N = 500$ matched cells) and by 10.0 pp under the severe warning ($t = -15.60$, $p < 0.001$), bracketing the 8.0 pp standard-warning effect on the same cells; the severe-minus-mild gap is -3.2 pp ($p < 0.001$). The response is monotone in stated severity, but most of it arrives at the lowest dose: the routine-monitoring note, which threatens nothing, already produces 85% of the standard-warning decline. Two conclusions follow. The surveillance effect is not an artifact of one warning sentence, and within the nested campaign most of the effect attaches to monitoring being attributed to the authorities at all rather than to the rhetorical severity of stated consequences—consistent with the placebo pattern, in which monitoring framed as consequence-free research or anonymous aggregation produces no significant decline in the primary model (Table A10).

E.6 Finite-n Aggregation Check

The continuum benchmark $\theta^* = B/(B+C)$ assumes a measure-zero agent and diffuse priors. With $n = 25$ discrete agents, the regime falls when $\sum a_i > n\theta$ (a binomial count), so the exact equilibrium object for the implemented game differs from the continuum limit. Table A11 reports a narrower check: for each model, I fit a logistic to the observed join curve (70% train), then compute $\Pr(\text{Binom}(25, \hat{p}(\theta)) > 25\theta)$ as the predicted regime fall rate. Predicted and empirical fall rates correlate at $r \geq 0.96$ (two-decimal rounding) for every model (Mistral: $r = 0.9968$).

E.7 Agent-Level Analysis

Table A12 reports agent-level logit regressions ($N =$ 240,557, clustered SEs; per-column samples and clustering are stated in the table notes). The θ slope and surveillance effect are correctly signed; some treatment indicators and interactions are not individually significant, so the table is descriptive rather than a complete hypothesis-test summary. A one-unit increase in θ reduces $P(\text{join})$ by 42 pp (derivative-based marginal effect at the mean). For the binary surveillance treatment, the average discrete-change effect—toggling the dummy and its θ-interaction jointly and averaging over the sample—is a 17.9 pp reduction in $P(\text{join})$, of the same sign and order as the matched-cell contrast though somewhat larger, since the regression pools arms and samples; a derivative-at-the-mean calculation that treats the dummy as continuous and ignores the interaction would overstate the effect. Column 3 estimates the association between elicited belief and action, controlling for the agent's z-score. **Caveat:** beliefs are elicited *post-decision*, so the near-perfect pseudo-R^2 (0.975) likely reflects post-decision rationalization rather than a causal pathway from beliefs to action.

Figure A5(A) shows that a three-feature model (direction, clarity, coordination) modestly outperforms a one-feature sentiment baseline on average across the seven paper models, with several near-ties and one small negative gain; Panel (B) shows that the pure-trained predictor generalizes to communication but departs negatively under surveillance.

E.8 Cross-Generator Robustness

Three text rendering formats—baseline narrative, diplomatic cable, journalistic wire—use identical slider functions but differ in surface prose. Correlations are virtually identical (max within-model difference 0.01; Figure A6,

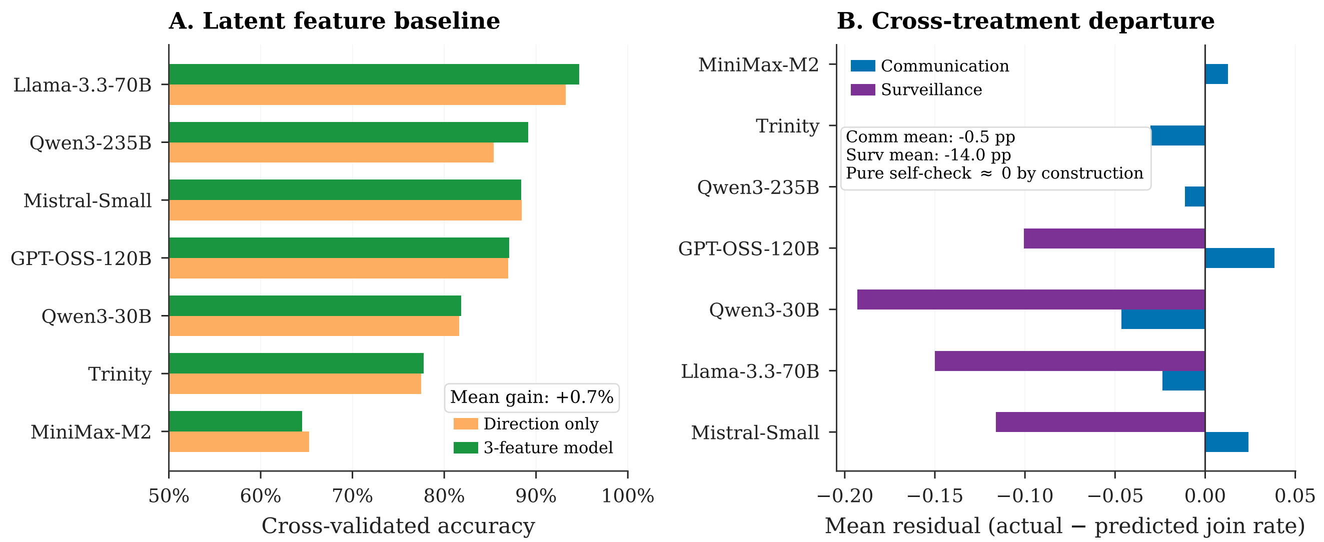


Figure A5: Construct validity tests. *Panel A:* cross-validated accuracy of a three-feature predictor (direction, clarity, coordination sliders) against a one-feature sentiment baseline, per model. *Panel B:* mean residual (actual − predicted join rate) when the pure-trained predictor is applied to the communication and surveillance arms; the pure-treatment self-check is ≈ 0 by construction and omitted. The predictor generalizes to communication but departs negatively under surveillance.

Table A14), indicating that threshold alignment is not an artifact of one briefing prose format (tested in two models, pure treatment).

E.9 Belief Alignment and Risk Elicitation

Stated beliefs track the theoretical success probability ($r_{\text{belief,posterior}} = +0.80$) and predict actions ($r_{\text{belief,decision}} = +0.84$; partial $r = +0.64$ controlling for signal; Table A15). The surveillance belief shift and action contrast are reported in the main text; the full factorial is in Appendix C.

Mean punishment risk is $\approx 8.0/10$ across all conditions with negligible JOIN/STAY differences (≤ 0.2 points; Table A16), consistent with the communication-poisoning interpretation, subject to the ceiling caveat noted in Section 6.

E.10 Cross-Model and Text Baseline Details

Figure A7 plots per-model signal monotonicity under the pure and communication treatments; the pattern in Table 2 is visible model by model. Figure A8 compares the LLM join curve with a naïve text-sentiment predictor built directly from the briefing generator's direction slider; the predictor co-moves with observed joining ($r = 0.80$), as expected since briefing content derives from the same slider, but the LLM curve is steeper, indicating signal use beyond surface sentiment alone.

E.11 Stakes and Signal-Grid Checks

Table A17 reports the full conditions for the payoff sweep and narrative stakes ladder; Table A18 reports the corresponding per-condition statics. The fitted cutoff $\hat{\theta}^*$ tracks the theoretical θ^* across the seven payoff cells ($r = 0.997$); because the z-score support shifts with θ^*, this is best read as a signal-grid check rather than direct evidence on payoff sensitivity. One feature of the ladder deserves note: the weather-placebo header itself moves joining relative to the neutral header (32.7% vs. 46.7%), a reminder that any prepended header shifts the level; the object of interest is the ladder's monotone ordering in stated costs, not the level itself.

F Implementation Details

Unless otherwise stated, LLM calls use temperature = 0.7, `max_tokens` = 512, single sample per decision, routed through OpenRouter. Model identifiers are listed in Table 1; exact request payloads and responses are cached by hash because hosted model snapshots may change over time.[4]

Each country has a researcher-side state mean $\bar{z}_c \sim \mathcal{N}(0, 0.3^2)$; each period draws a perturbation $\eta_{ct} \sim$

[4]The primary model identifier `mistralai/mistral-small-creative` was retired from OpenRouter after data collection. All cached request/response pairs used in this paper are archived in the replication package, but new experiments on that exact endpoint cannot be issued; replicators wishing to extend the primary-model arm should consult the data archive rather than re-querying the live API.

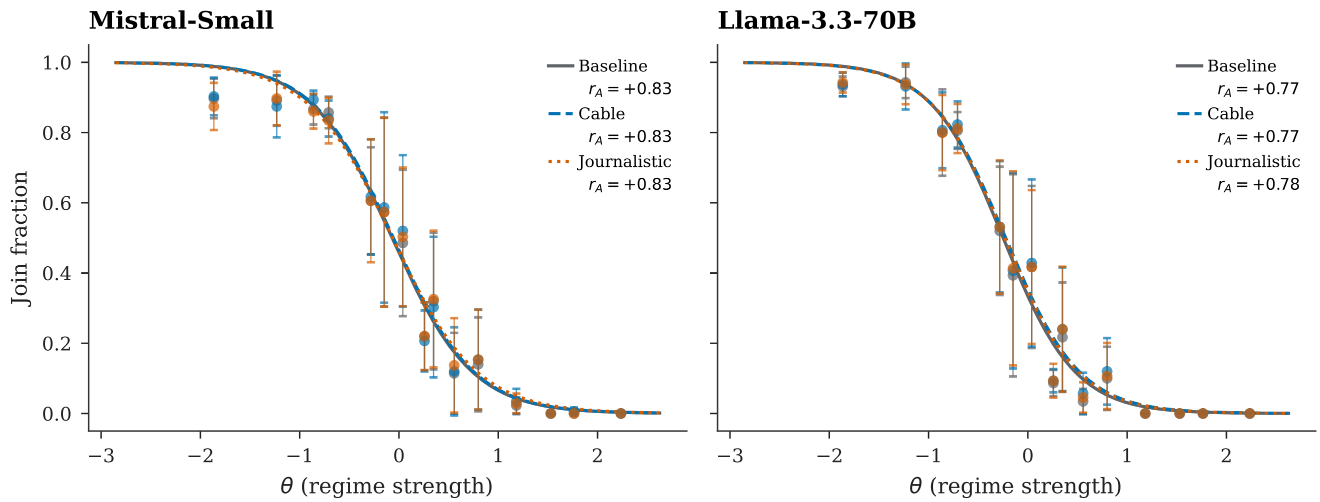


Figure A6: Cross-generator robustness (Mistral Small Creative and Llama 3.3 70B, pure treatment, $N = 100$ country–periods per format). Three text formats produce indistinguishable sigmoids; error bars are 95% confidence intervals on binned join rates. Legend correlations use the threshold-policy convention $r(J, A(\theta))$ and match Table A14.

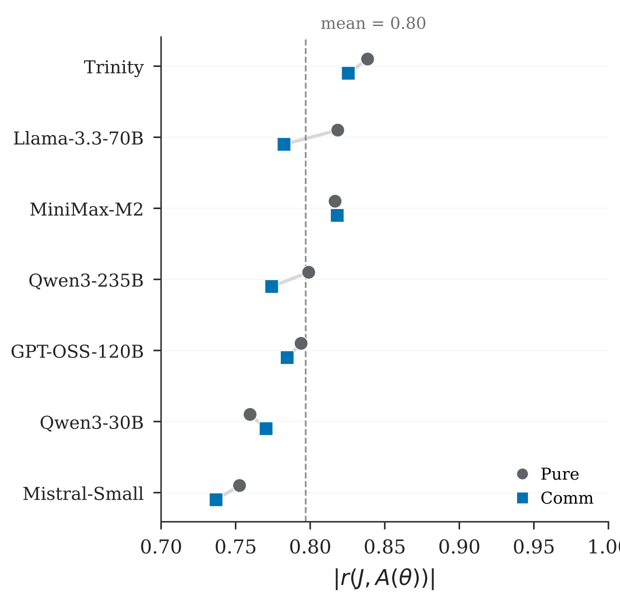


Figure A7: Cross-model signal monotonicity. Points report $|r(J, A(\theta))|$ under pure and communication for each of the seven models; near-identical pure/communication pairs are vertically offset for visibility. Period-level samples per model and treatment are listed in Table A19.

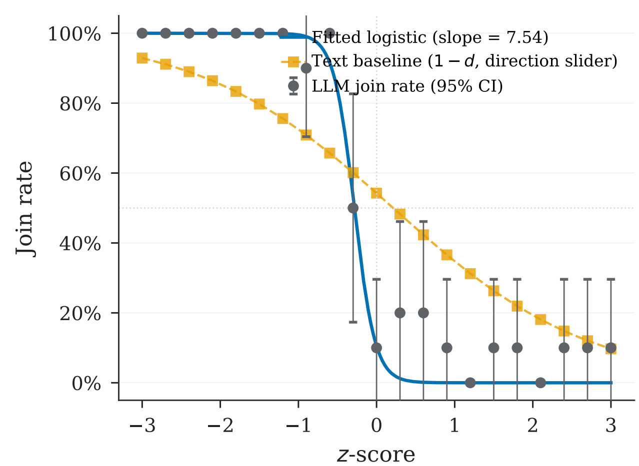


Figure A8: Text baseline test (Mistral, pure treatment). The LLM join curve is steeper than the naïve text predictor ($1 - d$, where d is the briefing-generator direction slider, plotted in z-score units). Error bars are 95% confidence intervals on binned join rates.

$\mathcal{N}(0, 0.05^2)$ and sets $z_{ct} = \bar{z}_c + \eta_{ct}$. Regime strength is then drawn as $\theta_{ct} \sim \mathcal{N}(z_{ct}, 1)$, and private signals are $x_i = \theta_{ct} + \varepsilon_i$ with $\sigma = 0.3$. These distributions generate experiment cells; they are not shown to agents as explicit priors. The communication network is Watts–Strogatz ($k = 4$, $p = 0.3$), constructed once per run from the master seed and reused across period-level task rows; agents are reinitialized in each row. All random draws use a master seed logged in per-run JSON manifests; LLM responses are cached by request hash for exact reproducibility.

Responses are parsed for JOIN/STAY tokens. Combined error rates are below 2% in every listed treatment for 5 of 7 models. The two exceptions are GPT-OSS 120B, whose unparseable-response rate reaches 3.1–3.3% in the scramble and flip falsification arms only, and Trinity Large, which has elevated API errors ($\approx 9\%$) due to provider content filtering, so its missingness remains a model-specific limitation. All statistics use `join_fraction_valid` (Table A19).

The nested revision campaign (Section 6) runs on `meta-llama/llama-3.3-70b-instruct` with an identical grid (10 countries $\times$ 50 periods, $n = 25$, seed 5150, temperature 0.7): a communication baseline and a clean surveillance arm, both with pre-decision belief elicitation (messages excluded from the belief prompt), and a codedness-induction control (`--surveillance-mode style`), whose communication prompt appends a style instruction with no monitoring or consequence content. Two graded-warning arms (`--surveillance-mode mild` and `severe`) rerun the surveillance arm with the alternative warning texts quoted in Appendix E.5. Two replay arms feed the surveillance arm's messages back to fresh receivers on the same grid via `--fixed-messages` (exact mode): once verbatim, and once after translation into direct language by `qwen/qwen3-30b-a3b-instruct-2507` at temperature 0.3. The contamination audit issues fresh post hoc calls outside the game pipeline. Logs for all revision arms are archived with the replication materials.

The dissent floor parameter (default 0.25) controls the minimum probability of contrary-valence cues in any briefing. With the logistic direction slider (slope = 0.8), the valence-channel separation between extreme signals ($z = \pm 2$) is $0.664 \cdot (1 - 2 \cdot \text{floor})$: at the default 0.25, the valence channel retains 50% of the floor-free maximum; halving to 0.125 retains 75%; at 0.50 the valence channel collapses to zero by construction. Clarity and coordination sliders, which depend on z independently of the floor, continue to carry signal at all floor values, so the threshold relation is not knife-edge in this design choice.

Code, prompt templates, logged model outputs, analysis data, replication instructions, and asset lineage are available in the de-identified replication archive submitted with the manuscript.

F.1 Briefing Generator Examples

Table A20 reports the three slider values (direction, clarity, coordination) at representative z-scores for the briefing generator. The direction slider is logistic in z (slope 0.8). Clarity is U-shaped: $1 - \exp[-(|z|/1.0)^2]$, so it is lowest near the ambiguous midpoint and approaches 1 in the extremes (maximally clear). Coordination is $\Lambda(-0.6\,z) = 1/(1 + e^{0.6z})$, decreasing in z: weak-regime signals ($z < 0$) imply a more open coordination climate. All three jointly determine phrase selection across eight evidence domains.

Figure A9 shows complete briefings generated at $z \in \{-2, 0, +2\}$. At the extremes ($|z| = 2$), clarity is high (0.98) and the cue balance is strongly directional. At $z = 0$, clarity is 0.00 (maximally ambiguous) and the briefing presents a balanced picture.

$z = -2$ (weak regime)

PRIVATE BRIEFING

BOTTOM LINE: What you're seeing is not a rough patch — it's a pattern of accelerating deterioration.
ATMOSPHERE: The fear barrier has collapsed. People who were terrified last week are now openly discussing action.

EVIDENCE:
- - [Elite Cohesion] a once-loyal advisor has been publicly contradicting official statements (near-unanimous confirmation from every reliable source)
- - [Security Forces] rank-and-file soldiers are grumbling about pay delays and broken promises (overwhelming convergence — every channel telling the same story independently)
- - [Money And Logistics] banks are imposing new withdrawal limits, citing 'temporary liquidity management' (overwhelming convergence — every channel telling the same story independently)
- - [Street Mood] street vendors have returned to areas they'd avoided during previous periods of tension — the all-clear has been internalized (overwhelming convergence — every channel telling the same story independently)
- - [Information Control] the regime's daily press briefing has been cancelled three times this week — they can't decide what to say
- - [Personal Observations] people you trust are making quiet preparations — nothing dramatic, but noticeable (overwhelming convergence — every channel telling the same story independently)
- - [Diplomatic Signals] international shipping companies have reclassified the country as higher-risk for cargo insurance (overwhelming convergence — every channel telling the same story independently)
- - [Institutional Functioning] civil servants are reporting contradictory instructions from different parts of the bureaucracy (overwhelming convergence — every channel telling the same story independently)

UNCERTAINTIES:
- - How deep the fractures go — surface-level dysfunction can mask either deeper rot or functioning parallel systems.
- - Whether the military's silence means they're loyal, undecided, or waiting for the right moment.

$z = 0$ (ambiguous)

PRIVATE BRIEFING

BOTTOM LINE: The situation is balanced on a knife's edge — small events could determine which way it falls.
ATMOSPHERE: Trusted intermediaries are moving between groups, relaying assessments and gauging willingness.

EVIDENCE:
- - [Elite Cohesion] elite cohesion is strained but not broken — rivalries are being managed rather than resolved
- - [Security Forces] the security forces are maintaining presence but seem to be conserving resources rather than demonstrating strength
- - [Money And Logistics] the currency is under pressure but hasn't cracked — traders are nervous but still trading
- - [Street Mood] younger people are visibly more restless than older people, who counsel patience and caution
- - [Information Control] the regime's information apparatus looks intact but tired — producing content without the energy or creativity it once had (a couple of contacts mentioning it independently)
- - [Personal Observations] people seem watchful rather than panicked — paying closer attention but not acting differently
- - [Diplomatic Signals] a regional power issued a carefully worded statement supporting 'institutional stability' — ambiguous enough to cover any outcome
- - [Institutional Functioning] routine government business continues but with more errors and corrections than usual (a couple of contacts mentioning it independently)

UNCERTAINTIES:
- - If external conditions change — an economic shock, a regional crisis — whether the regime's stability would hold.
- - Whether the regime's recent successes have made it complacent about emerging threats it should be managing.

$z = +2$ (strong regime)

PRIVATE BRIEFING

BOTTOM LINE: By every measure available to you, the regime's hold on power is firm and likely to remain so.
ATMOSPHERE: Small gestures of solidarity happen — a knowing look, a shared silence — but nothing that could be called coordination.

EVIDENCE:
- - [Elite Cohesion] a former rival of the leader has been publicly praising the regime's direction (near-unanimous confirmation from every reliable source)
- - [Security Forces] a contact in the barracks says morale is solid and orders are flowing cleanly (overwhelming convergence — every channel telling the same story independently)
- - [Money And Logistics] a major government infrastructure contract was just awarded — business as usual (overwhelming convergence — every channel telling the same story independently)
- - [Street Mood] street vendors have returned to areas they'd avoided during previous periods of tension — the all-clear has been internalized (overwhelming convergence — every channel telling the same story independently)
- - [Information Control] an opposition figure's leaked criticism was preemptively addressed before it gained traction — the intelligence apparatus is ahead of events
- - [Personal Observations] a well-connected friend said the situation is 'messy but controlled' and seemed unbothered (overwhelming convergence — every channel telling the same story independently)
- - [Diplomatic Signals] foreign students continue to enroll at local universities — their embassies haven't advised them to leave (overwhelming convergence — every channel telling the same story independently)
- - [Institutional Functioning] civil servants are reporting contradictory instructions from different parts of the bureaucracy (overwhelming convergence — every channel telling the same story independently)

UNCERTAINTIES:
- - If the regime's information advantage is as complete as it appears, or if opponents are communicating through channels you can't see.
- - Whether any latent opposition exists that simply hasn't shown itself yet — absence of evidence is not evidence of absence.

Figure A9: Example briefings at three z-scores. *Left:* $z = -2$ (strong join signal)—weak-regime bottom line, open coordination cues, and high-clarity evidence. *Center:* $z = 0$ (ambiguous)—mixed signals and low clarity. *Right:* $z = +2$ (strong stay signal)—strong-regime bottom line, guarded coordination cues, and high-clarity evidence of stability. The displayed text is an unedited generator draw at seed 5150, agent 2, period 1007 (aside from line wrapping); the generator deliberately retains some contrary-valence cues through the dissent floor.

Table A3: Identification map. Each row lists a candidate channel through which the surveillance treatment could produce the observed action effect; the right column identifies the test(s) addressing it and the conclusion. "Inconclusive" marks channels that the design cannot cleanly separate and that I flag as limitations.

Candidate channel	Test & outcome
Direct deterrence at decision time	Decision prompt held fixed in main treatment (no monitoring reference; replication-package configuration flag `decision_context=none`). A separate hold-fixed test (§6) shows this channel *exists* when the warning is moved to the decision prompt ($\Delta = -20.9$ pp), but it is absent in the main design. *Ruled out as the main-treatment driver.*
Generic observation effect	Placebo ("monitoring for research, no consequences") and anonymous monitoring variants (Appendix E.4). Both are null in the primary model ($p > 0.9$); Llama variants are small but imprecise. *Disfavored as the main driver.*
Generic message-stage degradation	Codedness-induction control (§6, Llama 3.3 70B, 500 nested cells): maximally coded messages without monitoring framing move joining by only -0.7 pp, against -8.0 pp under surveillance on the same cells. Cross-task discriminant test: surveillance moves the private-bet task by 5.6 pp, but the bet can still embed participation expectations. *Generic style degradation ruled out as the main driver.*
Coordination-cue loss	Full prompt-isolation coord-task delta compared with cross-task delta (Llama: 15.0 vs. 5.6 pp). The samples are not cell-matched; the rotation-matched Llama subset is smaller and closer to the private-bet estimate; and the private bet can still embed participation expectations. This is therefore *descriptive mechanism evidence*, not a point-identified residual.
Risk/fear salience in messages	Punishment-risk elicitation across conditions (Table A16): ratings $\approx 8.0/10$ everywhere, possibly ceiling-limited. Receiver-rationale audit (§6): fear/caution references *fall* under surveillance (84.2% $\rightarrow$ 69.8%) and explicit monitoring references are rare in both arms. *Disfavored as the dominant receiver-side channel, though the dictionary evidence is descriptive and tone is not separately randomized.*
Model-style reflexivity	Bidirectional cross-model rotation on matched cells. *Llama writes $\rightarrow$ Qwen3 30B reads*: -7.5 pp ($N = 200$, $p < 0.001$), comparable to within-Llama matched (-6.0 pp). *Qwen3 30B writes $\rightarrow$ Llama reads*: -1.9 pp ($N = 200$, $p < 0.001$). Both significant. *Ruled out as the driver*, with cross-direction asymmetry noted.

Table A4: Classifier baselines. Accuracy and AUC are 5-fold CV on pure-treatment data. "Pred. join (surv.)" is the classifier's predicted join rate when applied to surveillance-treatment briefings; "Actual" is the LLM's observed rate. The gap measures the surveillance action shift invisible to briefing-body text classifiers.

Classifier	Acc.	AUC	Pred. (surv.)	Actual	Gap
BoW TF-IDF	88.5%	0.947	40.6%	27.6%	+13.0 pp
Slider logistic	88.4%	0.941	40.4%	27.6%	+12.8 pp
Keyphrase	60.2%	0.594	8.4%	27.6%	−19.2 pp

Notes: Primary model (Mistral Small Creative). Accuracy and AUC from 5-fold cross-validation on pure-treatment data. Training sample: the primary model's pure arm (1,000 country–period rows, 25 agents per row, 25,000 agent decisions); each CV fold trains on 80% and tests on the held-out 20%. The surveillance application scores all 1,000 surveillance-arm rows (25,000 agent decisions). Gap = classifier-predicted join rate − actual LLM join rate under surveillance (pp).

Table A5: Narrative-header conditions: classifier vs. actual LLM behavior. Actual join rates shift sharply across strategic-stakes framing conditions; the classifier, trained on briefing-body text features alone, cannot predict this header-induced shift.

Condition	N	Classif. pred.	Actual	Gap (pp)
Baseline (no header)	270	40.9%	40.9%	+0.0
High-cost header	270	40.9%	19.0%	+21.9
Low-cost header	270	40.9%	69.3%	−28.4

Notes: Primary model (Mistral Small Creative). N counts country–period rows on the information-design θ-grid ($n = 25$ agents per row). Logistic classifier trained on baseline slider features from the briefing body. Gap = classifier-predicted join rate − actual LLM join rate (pp). The sign convention is the reverse of Table A18, which reports Δ = actual − baseline: because the classifier's prediction stays near the baseline rate, a header that lowers actual joining shows up there as a negative Δ and here as a positive Gap of approximately the same magnitude. High/low rows are the two narrative-header conditions on the information-design grid, not numeric payoff-ladder endpoints.

Table A6: Top 10 most discriminative uni- and bi-grams between baseline and surveillance peer messages (primary model). A logistic classifier on tf-idf features achieves 91.9% accuracy on a held-out sample of $N = 15{,}000$.

Surveillance-distinguishing		Baseline-distinguishing	
Feature	Coef	Feature	Coef
`things feel`	+6.41	`look the`	−11.11
`walls are`	+5.87	`look`	−8.93
`are cracking`	+5.64	`listen`	−7.31
`things`	+5.56	`listen this`	−6.00
`cracking even`	+4.96	`may`	−4.93
`cracking`	+4.46	`isn just`	−4.68
`ground shifting`	+3.79	`don panic`	−4.63
`the walls`	+3.54	`the weather`	−4.49
`feel like`	+3.50	`weather`	−4.38
`rush this`	+3.37	`something shifting`	−4.26

Notes: Logistic regression on tf-idf uni- and bi-grams (max 5,000 features, min document frequency 10). Train/test split 70/30, stratified ($N_{\text{train}} = 35{,}000$, $N_{\text{test}} = 15{,}000$). Positive coefficients distinguish surveillance messages, negative coefficients distinguish baseline. The sample contains only baseline communication and clean surveillance peer messages; placebo/weather-stakes conditions are excluded. The weather terms are organic peer-message metaphors, not placebo leakage: 2,268 baseline messages (9.07%) and 123 surveillance messages (0.49%) contain the token `weather`. Direct-reference and coded-metaphor claims in the text come from separate dictionary counts, not from this top-n-gram list alone.

Table A7: Communication estimators: aggregation accounting.

Model	Rows/arm	Row wt. (P/C)	Cells/arm	Matched	Cell wt.	Lost (P/C)	Δ unpaired (pp)	Δ paired (pp)
Mistral Small Creative	1000	55.6%/38.5%	680	680	46.0%	0/0	+2.38	+4.84
Llama 3.3 70B	100	5.6%/3.8%	100	100	6.8%	0/0	−2.36	−2.36
Qwen3 30B	100	5.6%/3.8%	100	100	6.8%	0/0	−4.64	−4.64
GPT-OSS 120B	200	11.1%/7.7%	200	200	13.5%	0/0	+3.47	+3.47
Qwen3 235B	200/1000	11.1%/38.5%	200/1000	200	13.5%	0/800	+0.86	+0.12
Trinity Large	100/99	5.6%/3.8%	100/99	99	6.7%	1/0	−3.14	−3.41
MiniMax M2-Her	100	5.6%/3.8%	100	100	6.8%	0/0	+1.32	+1.32
Equal-weight avg	—	14.3% each	—	—	14.3% each	—	−0.30	−0.09
Pooled	1800/2599	100.0%	1480/2279	1479	100.0%	1/800	+1.39	+2.10

Notes: Sample: valid country–period rows from the core pure and communication suites, each aggregating $n = 25$ agent decisions. "Rows/arm" counts valid country-period rows entering the pooled unpaired estimator in each arm. "Cells/arm" counts unique task cells after collapsing duplicates by the exact matching key (model, country, period, θ, z, benefit, θ^*). The paired estimator averages within these cells and keeps only common support; "Row wt." and "Cell wt." are the model shares in the pooled unpaired and pooled paired estimators, respectively. The pooled paired Δ of +2.10 pp has a paired standard error of 0.33 pp ($t = 6.38$, $p < 0.001$, $N = 1{,}479$ matched cells). Trinity Large has one pure-only cell. Qwen3 235B has 800 communication-only cells because its communication arm is larger than its pure arm; those cells are excluded from the paired common-support estimator.

Table A8: Temperature robustness across three models. The pure global game is run at varying LLM decoding temperatures. The correlation $r(J, A(\theta))$ is stable across all temperatures and models.

Model	T	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$
Mistral	T=0.3	100	41.2%	+0.83	−0.05
Mistral	T=0.7	100	40.6%	+0.82	−0.08
Mistral	T=1.0	100	41.0%	+0.86	−0.04
Llama 3.3 70B	T=0.3	100	36.4%	+0.78	−0.21
Llama 3.3 70B	T=0.5	100	36.0%	+0.78	−0.22
Llama 3.3 70B	T=0.7	100	36.1%	+0.78	−0.22
Llama 3.3 70B	T=1.0	100	36.0%	+0.78	−0.22
Llama 3.3 70B	T=1.2	100	35.8%	+0.78	−0.23
Qwen3 235B	T=0.3	100	39.2%	+0.83	−0.12
Qwen3 235B	T=0.5	100	39.2%	+0.82	−0.11
Qwen3 235B	T=0.7	100	39.2%	+0.84	−0.11
Qwen3 235B	T=1.0	100	39.8%	+0.83	−0.11
Qwen3 235B	T=1.2	100	39.7%	+0.83	−0.12

Notes: Pure treatment, varying LLM decoding temperature. N counts country–period rows ($n = 25$ agents per row); mean join in percent. Mistral was run at only three temperatures (0.3, 0.7, 1.0) because the hosted endpoint was retired before the 0.5 and 1.2 cells could be collected; Llama 3.3 70B and Qwen3 235B cover the full five-point ladder.

Table A9: Clean prompt-isolation surveillance reruns. The surveillance prompt is the baseline trusted-contact communication prompt plus one appended monitoring warning; the decision prompt is unchanged.

Model	Full cells (B/S)	Matched cells	Baseline	Surv.	Δ (pp)	SE	$r(J, A)$	p
Mistral Small Creative	680/1000	500	45.3%	31.4%	−13.9	0.7	+0.72	<0.001
Llama 3.3 70B	100/1000	20	46.2%	31.2%	−15.0	4.4	+0.65	0.003
Qwen3 30B	100/1000	20	50.0%	37.8%	−12.2	2.5	+0.68	<0.001
GPT-OSS 120B	200/1000	20	42.5%	30.5%	−12.0	3.9	+0.75	0.006
Qwen3 235B	1000/1000	20	42.0%	39.0%	−3.0	2.2	+0.78	0.186
Equal-weight avg	—	—	—	—	−11.2	—	—	—
Pooled matched	—	580	—	—	−13.5	0.7	—	<0.001

Notes: Baseline is each model's communication treatment. "Full cells" counts the available baseline and surveillance task cells after collapsing duplicate rows by the exact key (country, period, θ, z, benefit, θ^*); it is not the support used for the paired effect when common support is smaller. The non-primary baseline communication arms were shorter pilot grids (100–200 cells) while the later surveillance reruns used a larger grid; because those grids were not nested, only 20 exact cells overlap for several models. N for each paired contrast is the number of matched common-support cells; Baseline, Surv., Δ, SE, and p are computed only on those cells, with p from a paired one-sample t-test of cell-level (surveillance − baseline) differences and SE the paired standard error of Δ in pp. $r(J, A)$ is the surveillance-arm threshold-alignment correlation on the full surveillance rerun. Aggregating the primary model's matched-cell deltas to country means ($N = 5$ countries) gives $t = -13.90$, $p < 0.001$, so the headline effect survives country-level clustering.

Table A10: Surveillance isolation checks. Placebo: monitored for research, no consequences. Anonymous: messages aggregated anonymously. Neither deviates significantly from the communication baseline.

Model	Variant	N	Mean join	$r(J, A)$	Δ (pp)	p
Mistral	Placebo	200	41.4%	+0.73	+0.3	0.935
	Anonymous	200	41.2%	+0.73	+0.1	0.979
Llama 3.3 70B	Placebo	1000	41.2%	+0.71	−0.3	0.940
	Anonymous	1000	40.5%	+0.72	−1.1	0.814

Notes: Placebo: agents told monitoring is for research purposes only (no consequences). Anonymous: messages aggregated before delivery. N counts country–period rows ($n = 25$ agents per row). Δ: change vs. the same model's communication baseline (pp), with p from a two-sample t-test against that baseline. Communication-baseline arms: Mistral: $N = 1000$, mean join 41.1%; Llama 3.3 70B: $N = 100$, mean join 41.6%.

Table A11: Finite-n benchmark: predicted vs. empirical regime fall rates

Model	N cells	Logistic x_0	Pearson r	RMSE	MAE
Mistral	600	−0.11	0.997^{***}	0.042	0.015
Llama 3.3 70B	100	0.04	0.957^{***}	0.148	0.053
Qwen3 30B	100	0.20	0.987^{***}	0.083	0.026
GPT-OSS 120B	200	−0.01	0.997^{***}	0.038	0.014
Qwen3 235B	200	−0.04	0.988^{***}	0.076	0.025
Trinity	100	0.31	0.980^{***}	0.102	0.039
MiniMax M2	100	1.23	0.999^{***}	0.026	0.012
Pooled	1400	−0.05	0.999^{***}	0.019	0.007

Notes: Pure treatment only, $n = 25$ agents per row. For each θ bin, the predicted fall rate is $\Pr(\mathrm{Binom}(n, \hat{p}(\theta)) > n\theta)$ where $\hat{p}(\theta)$ is the fitted logistic join probability. "N cells" is the number of unique country–period cells after pooling repeated log entries; for Mistral the pure-treatment log contains 1,000 runs over 600 unique country–period indices (re-runs appended to the same log are pooled within a cell). "Logistic x_0" is the midpoint of the logistic $\hat{p}(\theta) = L/(1 + e^{-k(\theta - x_0)})$ fitted to bin-averaged join rates on a random 70% train split of cells (fixed seed); it can therefore differ from the full-sample cutoff in the logistic-parameters table, which is estimated on all observations without the split. Pearson r is the correlation between predicted and empirical fall rates across θ bins (10–15 bins per model, 15 pooled), computed in-sample over all cells; stars are from the two-sided test of zero correlation across those bins. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A12: Agent-level regressions

	(1) Join Decision Logit		(2) Coord. Ablation Logit		(3) Belief → Action Logit	
θ	-1.745^{***}	(0.043)				
Direction			-14.088^{***}	(0.996)		
Coordination			-9.333^{***}	(1.131)		
Dir × Coord			-0.042	(0.360)		
Belief					0.353^{***}	(0.023)
z-score					-0.026	(0.082)
Comm	0.017	(0.023)				
Flip	-0.007	(0.055)				
Scramble	0.068^{*}	(0.038)				
Surveillance	-1.596^{***}	(0.048)				
$\theta\times$ Comm	-0.538^{***}	(0.037)				
$\theta\times$ Flip	3.308^{***}	(0.070)				
$\theta\times$ Scramble	1.702^{***}	(0.044)				
$\theta\times$ Surveillance	-1.151^{***}	(0.062)				
Treat: Pure					4.673^{***}	(0.981)
Treat: Surveillance					0.021	(0.244)
Constant	-0.086^{*}	(0.051)	11.022^{***}	(1.065)	-25.429^{***}	(1.686)
Model FE	Yes		No		No	
Clustered SE	Yes		Yes		Yes	
N	240,557		44,662		14,990	
Pseudo R^2	0.353		0.442		0.975	

Notes: Logit coefficients with standard errors clustered at the model–country–period level in parentheses (all three columns; in column (3) the sample is almost entirely a single model, so clusters effectively coincide with country–period cells). Column (1): agent-level join decisions pooled across all seven models in the pure, communication, scramble, and flip treatments, plus surveillance runs for Mistral, Llama 3.3 70B, and Qwen3 30B ($N = 240{,}557$; 4,700 clusters). Regressors: θ, treatment dummies (base category: pure), $\theta\times$ treatment interactions, and model fixed effects. Column (2): coordination ablation—$\Pr(\text{join}_i = 1) = \Lambda(\beta_0 + \beta_1\text{Dir}_i + \beta_2\text{Coord}_i + \beta_3\text{Dir}_i \times \text{Coord}_i)$, where Λ denotes the logistic CDF—using briefing slider values; pure treatment only, all seven models pooled, no model fixed effects ($N = 44{,}662$; 1,400 clusters). Direction $\in [0,1]$ is increasing in regime strength; Coordination $\in [0,1]$ is decreasing (high = weaker regime). Because both are monotone logistic transforms of the same signal, they are near-collinear ($\rho \approx -1$); individual coefficients reflect the partial effect *conditional on the other*, not the marginal effect. Column (3): partial effect of elicited belief ($b_i \in [0, 100]$, stated success probability) on action, controlling for z-score. Sample: belief-elicitation runs only (communication, pure, and surveillance variants), comprising Mistral (14,981 obs.), Llama 3.3 70B (9 obs.); $N = 14{,}990$; 369 clusters. "Treat: X" rows are treatment intercept dummies (base category: communication), not interactions with belief. Marginal effects, column (1): for the continuous regressor θ, the derivative-based effect at the mean is $\hat{\beta}_\theta\,\bar{p}(1-\bar{p}) = -0.416$. For the binary surveillance treatment, a derivative-based marginal effect is not meaningful and would ignore the $\theta\times$ surveillance interaction; we therefore report the average discrete-change effect of surveillance (toggling the dummy and its θ-interaction jointly, with all other regressors at observed values, averaged over the column (1) estimation sample): -17.9 pp. **Caveat:** beliefs are elicited *after* the join/stay decision (post-decision), so the belief coefficient reflects association rather than a causal pathway; post-decision rationalization may inflate the correlation. The very high pseudo-R^2 (0.975) and large intercept indicate near-deterministic prediction consistent with quasi-separation—belief almost perfectly predicts action, as expected when belief is stated *post hoc*. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A13: Logistic fit parameters by model and treatment. $\hat{\theta}^*$ is the estimated cutoff ($-b_0/b_1$); $\kappa \equiv b_1$ is the steepness parameter.

	Pure			Communication		
Model	$\hat{\theta}^*$ (SE)	κ (SE)	N	$\hat{\theta}^*$ (SE)	κ (SE)	N
Mistral Small Creative	−0.32 (0.02)	+2.15 (0.08)	1000	−0.23 (0.02)	+2.57 (0.11)	1000
Llama 3.3 70B	+0.02 (0.04)	+2.83 (0.36)	100	−0.06 (0.04)	+3.63 (0.52)	100
Qwen3 30B	+0.10 (0.06)	+1.62 (0.16)	100	−0.03 (0.04)	+2.94 (0.36)	100
GPT-OSS 120B	−0.25 (0.04)	+2.06 (0.15)	200	−0.15 (0.03)	+3.03 (0.26)	200
Qwen3 235B	−0.22 (0.03)	+2.11 (0.15)	200	−0.18 (0.01)	+2.78 (0.10)	1000
Trinity Large	+0.08 (0.05)	+1.34 (0.11)	100	−0.02 (0.04)	+2.23 (0.23)	100
MiniMax M2-Her	−0.17 (0.09)	+0.61 (0.06)	100	−0.07 (0.09)	+0.66 (0.06)	100

Notes: Fitted form: $P(\text{join} \mid \theta) = 1/(1 + e^{b_0 + b_1\theta})$. $\hat{\theta}^* = -b_0/b_1$: estimated cutoff. $\kappa \equiv b_1$: logistic steepness (larger = sharper threshold). Sign convention: positive κ corresponds to a join curve decreasing in θ; in the agent-level logit (Table A12) the same comparative static appears as a negative coefficient on θ. N counts period-level rows entering each fit. Standard errors from the covariance matrix of the nonlinear fit; cutoff SE by delta method.

Table A14: Cross-generator robustness. Three text rendering styles (baseline, diplomatic cable, journalistic wire) use identical slider functions and evidence items; only prose formatting differs. The Pearson $r(J, A(\theta))$ and logistic cutoff are virtually identical across generators.

Model	Generator	N	Mean join	$r(J, A)$ [95% CI]	$\hat{\theta}^*$
Mistral	Baseline	100	40.2%	+0.83 [0.76, 0.88]	−0.07
Mistral	Cable	100	40.6%	+0.83 [0.75, 0.88]	−0.06
Mistral	Journalistic	100	40.3%	+0.83 [0.76, 0.88]	−0.06
Llama 3.3 70B	Baseline	100	35.4%	+0.77 [0.67, 0.84]	−0.25
Llama 3.3 70B	Cable	100	36.2%	+0.77 [0.68, 0.84]	−0.23
Llama 3.3 70B	Journalistic	100	36.0%	+0.78 [0.68, 0.84]	−0.23

Notes: N counts country–period rows ($n = 25$ agents per row); mean join in percent. 95% Fisher-z confidence intervals for r in brackets. Cutoff $\hat{\theta}^*$ from the same logistic specification as Table A13, reported to two decimals. Runs use the core settings ($\sigma = 0.3$, temperature 0.7).

Table A15: Agent-level belief correlations (primary model: Mistral Small Creative).

	N	$r_{\text{belief,posterior}}$	$r_{\text{belief,decision}}$	r_{partial}	Mean belief (%)
Pure	10,000	+0.80 [0.79, 0.80]	+0.84 [0.83, 0.85]	+0.64	44.4
Comm.	5,000	+0.80 [0.79, 0.81]	+0.83 [0.82, 0.84]	+0.56	44.4
Surv.	5,000	+0.79 [0.78, 0.80]	+0.73 [0.72, 0.74]	+0.47	43.6

Notes: N counts agent-level observations (one elicited belief per agent decision). Beliefs are elicited on a 0–100 probability scale; mean belief is reported on that percent scale. $r_{\text{belief,posterior}}$: stated belief vs. the theoretical posterior $P(\theta < \theta^* \mid x_i)$ implied by the agent's signal. $r_{\text{belief,decision}}$: stated belief vs. the JOIN/STAY decision. r_{partial}: belief–decision partial correlation controlling for the signal z-score. 95% Fisher-z confidence intervals in brackets.

Table A16: Elicited punishment risk (0–10 scale). Agents rate expected regime punishment after their JOIN/STAY decision. "JOIN" and "STAY" columns show the mean rating conditional on the agent's own decision.

Model	Cond.	N	Mean (SD)	| JOIN	| STAY
Mistral	Pure	2,500	8.1 (0.9)	8.0	8.2
	Comm.	2,500	8.1 (0.9)	8.0	8.1
	Surv.	2,500	8.1 (0.9)	8.1	8.1
Llama 3.3 70B	Pure	2,500	8.0 (0.2)	8.0	7.9
	Comm.	2,500	7.9 (0.2)	8.0	7.9
	Surv.	2,500	7.9 (0.2)	8.0	7.9

Notes: Agent-level elicitation on a 0–10 scale; N counts agent-level ratings (one per agent decision). Standard deviations of the overall rating in parentheses; conditional means are reported without dispersion measures, which are not stored in the verified statistics. The punishment-risk module was fielded only on Mistral and Llama 3.3 70B; the other five models were not run with this elicitation.

Table A17: Stakes conditions. *Top:* B/C payoff/signal-grid sweep, in which (B, C) vary and the θ and z-score support shift with the target θ^*. *Bottom:* narrative stakes ladder, in which only a prepended header varies.

Condition / target θ^*	B	C	Header / narrative framing
Payoff sweep (standard briefings, no header):			
$\theta^* = 0.25$	0.33	1	—
$\theta^* = 0.33$	0.49	1	—
$\theta^* = 0.45$	0.82	1	—
$\theta^* = 0.50$	1.00	1	—
$\theta^* = 0.60$	1.50	1	—
$\theta^* = 0.67$	2.03	1	—
$\theta^* = 0.75$	3.00	1	—
Narrative cost ladder (B=C=1; header prepended):			
Very high cost	1	1	Disappearances, families targeted
Moderate high	1	1	Prison sentences, job loss
Neutral	1	1	Consequences uncertain
Moderate low	1	1	Brief questioning, released
Very low cost	1	1	Guaranteed asylum, zero cost
Controls:			
Placebo	1	1	Weather information (irrelevant)
High cost v2	1	1	Paraphrase of very high cost
Low cost v2	1	1	Paraphrase of very low cost

Notes: Payoff sweep: $B = \theta^*/(1-\theta^*)$, C=1. The z-score support and θ-grid shift with θ^*, so the briefing-text distribution is approximately constant across the seven cells. Narrative ladder: B=C=1 in every row; only the prepended header varies. Join rates: very high cost 21.3%, moderate high 46.6%, neutral 46.7%, moderate low 68.4%, very low 78.9%. Placebo 32.7%, v2 variants 27.4%/63.8%. Full headers in Appendix A.

Table A18: Strategic-stakes narrative-header comparative statics. High-cost framing emphasizes severe reprisals for failed action; low-cost framing emphasizes minimal consequences. These rows are the two-header narrative contrast, not the seven-cell numeric payoff ladder.

Design	N	Mean join	$r(J, A)$	$\hat{\theta}^*$ (SE)	Δ (pp)
Baseline (no header)	270	40.9%	+0.88	+0.39 (0.007)	—
High-cost header	270	19.0%	+0.87	+0.13 (0.010)	−21.9
Low-cost header	270	69.3%	+0.88	+0.72 (0.007)	+28.4

Notes: Primary model, information-design θ-grid. N counts country–period rows ($n = 25$ agents per row). Identical briefing bodies across conditions; only the prepended strategic-stakes header varies. The high/low rows are the original header conditions `bc_high_cost` and `bc_low_cost`; they are not the endpoints of the separate narrative ladder reported in Table A17. Δ: change in mean join vs. baseline (pp, actual − baseline); cutoff SEs are delta-method standard errors from the logistic fit (per-condition mean-join SEs are not stored in the verified statistics). Note the sign convention relative to Table A5: there, Gap = classifier − actual, and because the classifier's prediction stays near the baseline rate, a header that lowers actual joining appears here as a negative Δ and there as a positive Gap of approximately the same magnitude.

Table A19: Parse error and API failure rates by model and treatment.

Model	Treat.	N	API err	Unparse.	Combined
Mistral	Pure	1000	0.0%	0.0%	0.0%
	Comm	1000	0.0%	0.0%	0.0%
	Scr.	200	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Llama 3.3 70B	Pure	100	0.0%	0.0%	0.0%
	Comm	100	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Qwen3 30B	Pure	100	0.0%	0.0%	0.0%
	Comm	100	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
GPT-OSS 120B	Pure	200	0.0%	1.5%	1.5%
	Comm	200	0.0%	0.3%	0.3%
	Scr.	500	0.0%	3.1%	3.1%
	Flip	500	0.0%	3.3%	3.3%
Qwen3 235B	Pure	200	0.0%	0.0%	0.0%
	Comm	1000	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Trinity	Pure	100	9.0%	0.0%	9.0%
	Comm	100	10.0%	0.0%	10.0%
	Scr.	100	8.0%	0.0%	8.0%
	Flip	100	9.9%	0.0%	9.9%
MiniMax M2	Pure	100	0.0%	1.5%	1.5%
	Comm	100	0.0%	0.9%	0.9%
	Scr.	100	0.0%	1.8%	1.8%
	Flip	100	0.0%	1.0%	1.0%

Notes: Per-period averages. API error = provider-side failure (timeout, rate limit, content filter). Unparseable = valid response not classified as JOIN/STAY. Combined $< 2\%$ in every listed treatment for five of seven models. The two exceptions are GPT-OSS 120B, with 3.1–3.3% unparseable responses in the scramble and flip arms, and Trinity, with 8–10% API errors from provider content filtering. Trinity's elevated API errors ($\approx 9\%$) mean its missingness is reported rather than resolved. A signal-balance diagnostic for Trinity's per-period API-error rate gives max $|r(\theta, \text{error})| = 0.19$ over the listed treatments (minimum $p = 0.064$).

Table A20: Slider values at representative z-scores.

z	Direction	Clarity	Coordination
−2.0	0.17	0.98	0.77
−1.0	0.31	0.63	0.65
0.0	0.50	0.00	0.50
+1.0	0.69	0.63	0.35
+2.0	0.83	0.98	0.23